



Chapter 4 — Differential Amplifiers

Design of Analog CMOS
Integrated circuits - Behzad
Razavi

Presented by - Chinmay Joshi



CHAPTER OVERVIEW

Concepts covered -

1. Single Ended and Differential Operation
2. Basic Differential Pair
 - a. Qualitative Analysis
 - b. Quantitative Analysis
3. Common Mode Response
4. Differential Pair with MOS Loads



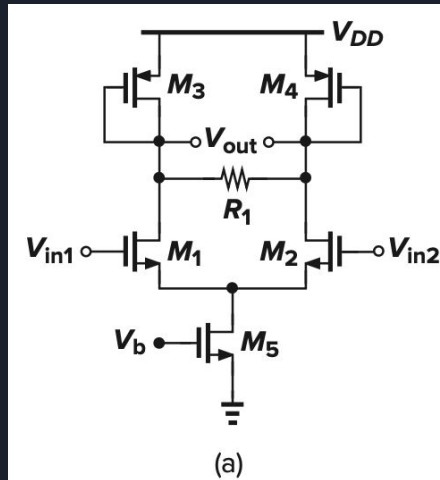
Additional Points

- Cadence Virtuoso/LTSpice is used for schematic design and circuit simulation.
- The GPDK180 technology library is used for all circuit simulations.
- A supply voltage of $VDD = 1.8V$ is used for both analytical calculations and simulations based on the recommended operating conditions of the GPDK180 technology model.
- The textbook considers $VDD = 3V$; therefore, slight differences may be observed between textbook results and the obtained results due to the difference in supply voltage.

Problem 4.16-a

4.16. Assuming that all the circuits shown in Fig. 4.44 are symmetric, sketch V_{out} as **(a)** V_{in1} and V_{in2} vary differentially from zero to V_{DD} , and **(b)** V_{in1} and V_{in2} are equal and vary from zero to V_{DD} .

Diagram -



Handwritten Solution

4-16 Ya9

a-y

For part 'a' when $V_{in1} - V_{in2}$ varies from 0 to V_{DD}

$V_{in1} - V_{in2} = V_{in,DM}$

As $V_{in,DM}$ increases

$I_{O1} \gg I_{O2}$

V_{out} decreases [$V_{out} = V_{out1} - V_{out2}$], as $V_{in,DM}$ ↑

I_{O1} ↑, ∴ drop in left branch increases

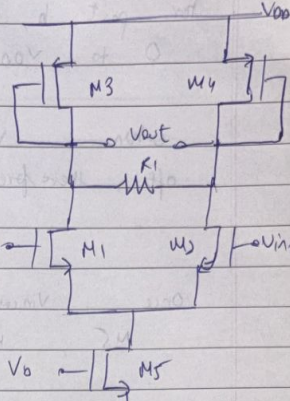
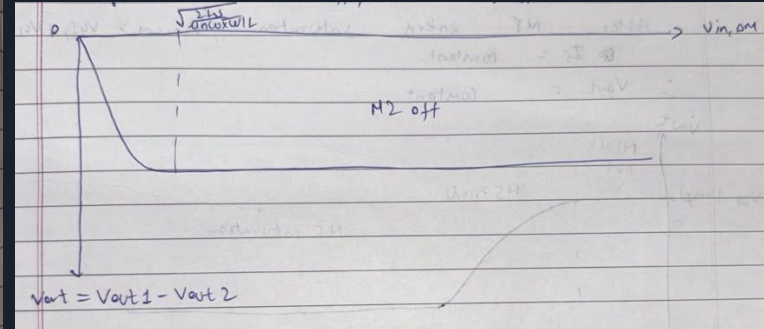
V_{out1} decreases,

after $V_{in,DM} = \sqrt{2} V_{Ox,eqilibrium}$, $M2$ turns off

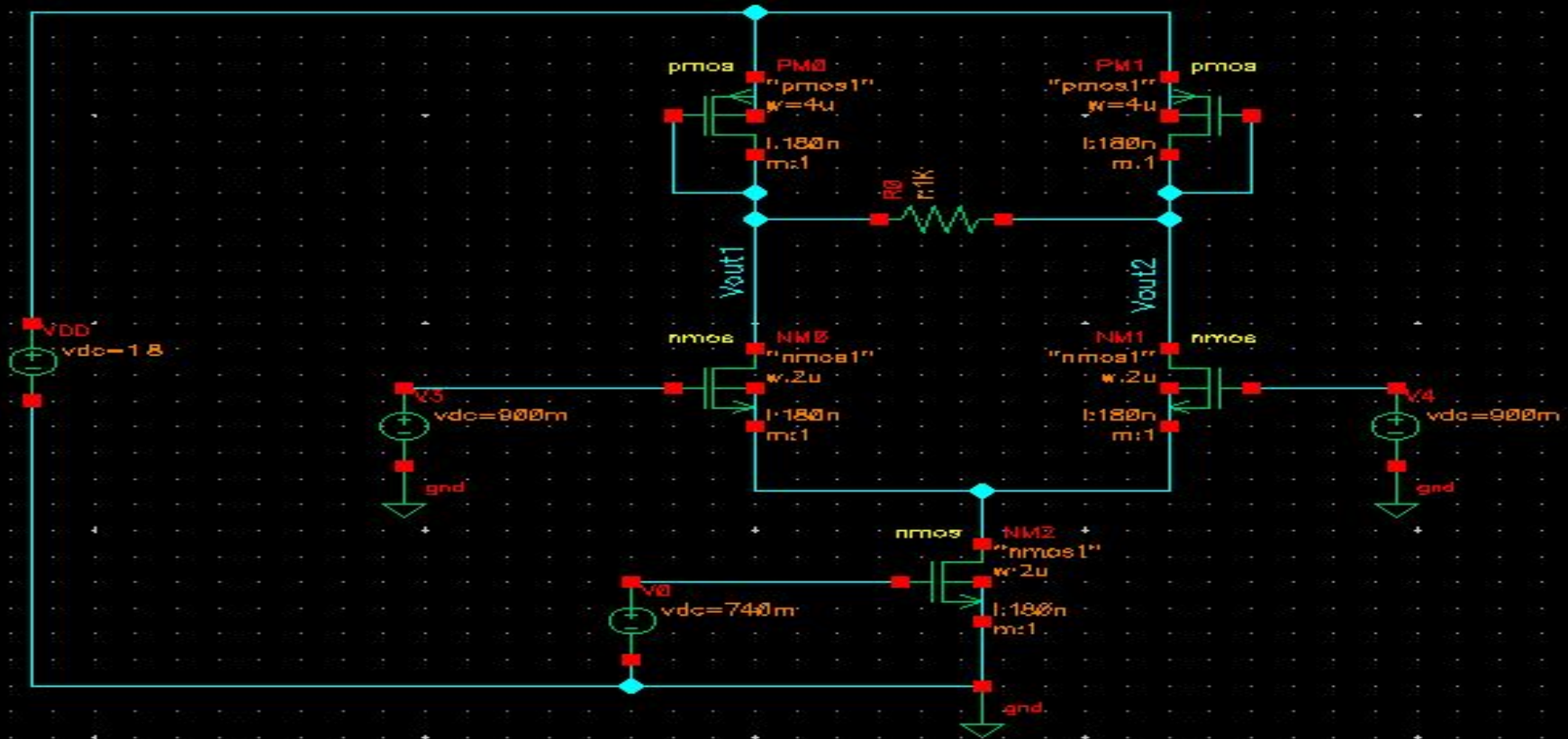
$M2$ turns off when $V_{GS2} = V_{th}$, $V_{DS2} = V_{th}$

∴ $M2$ turns off when $V_{in,DM} = \sqrt{\frac{2 I_{SS}}{K_2 \mu_n C_{ox} W/L}}$

after $M2$ is off, V_{out} becomes constant

Cadence Virtuoso Schematic



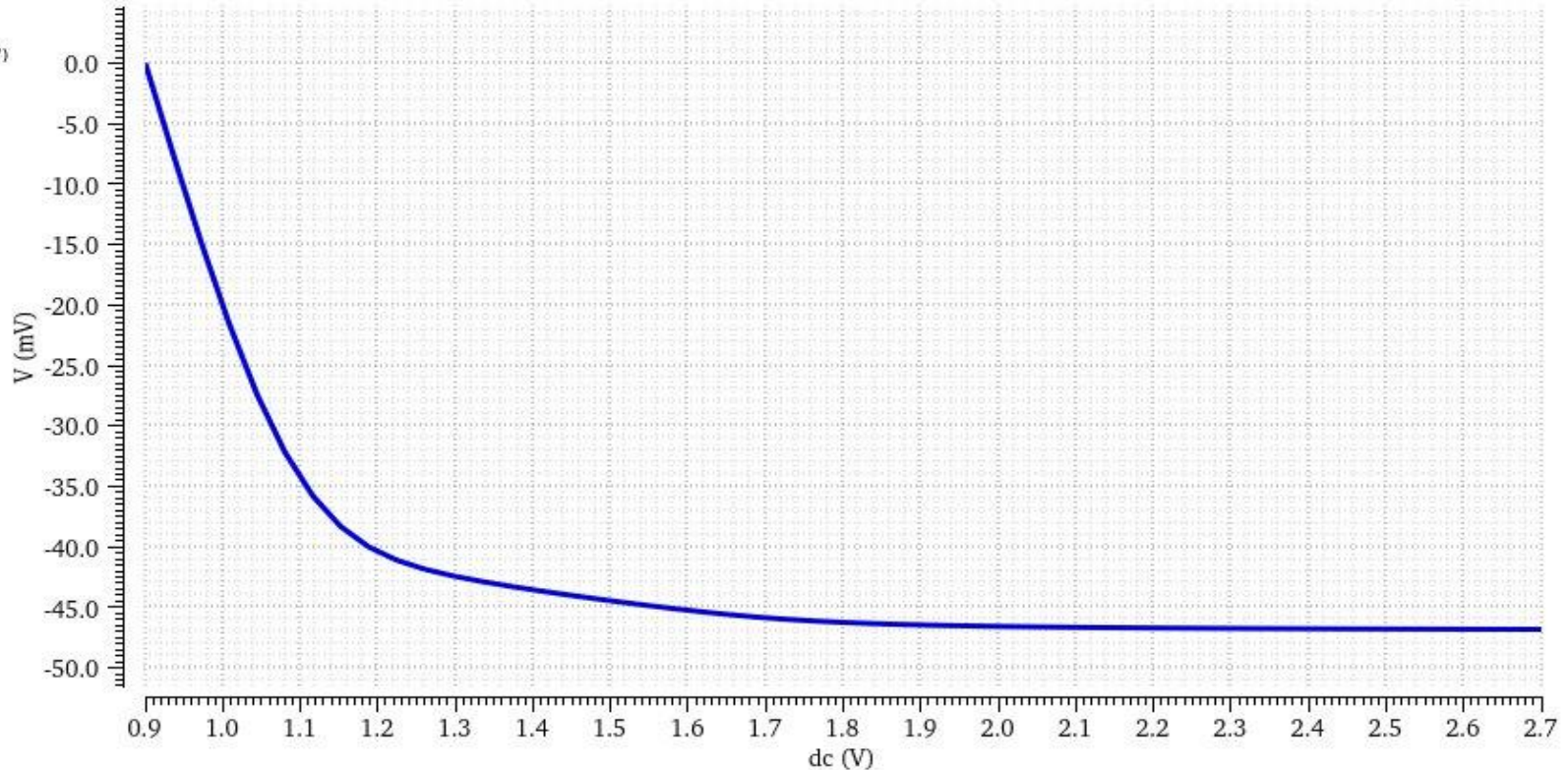
Cadence ADE-L DC Simulation for Vout

DC Response

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Name

...-v("/Vout2" ?result "dc")



Handwritten Solution

For part b, $V_{in1} = V_{in2} = V_{in,cm}$ varies from 0 to V_{DD}

When $V_{in,cm} < V_{th}$, $M1$ and $M2$ are off, therefore

$$V_{out} = V_{DD} - (V_{thp})$$

Once $V_{in,cm} > V_{th}$, current depends on $M5$ boundary condition for $M5$ is

$$V_{ds,s} = V_{gs,s} - V_{th}$$

for $V_{ds,s} < V_{gs,s} - V_{th}$, $M5$ is in triode

$$V_{in} - V_{ds,s} < V_{gs,s} - V_{th}$$

$$\therefore V_{in,cm} < V_{gs,s} + V_{ds,s} - V_{th}$$

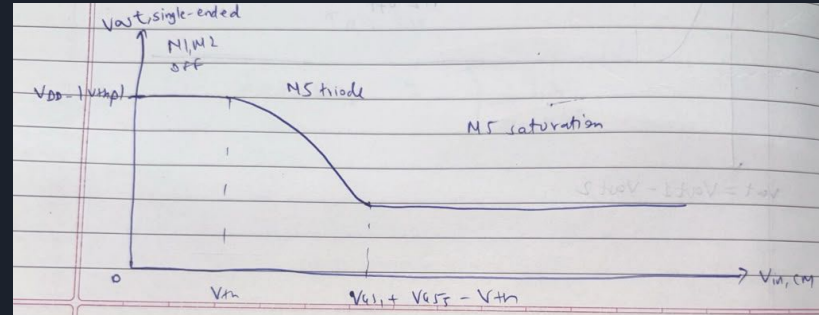
As $V_{in,cm}$ increases, current increases

V_{out} decreases

After $M5$ enters saturation, $V_{in,cm} > V_{gs,s} + V_{ds,s} - V_{th}$

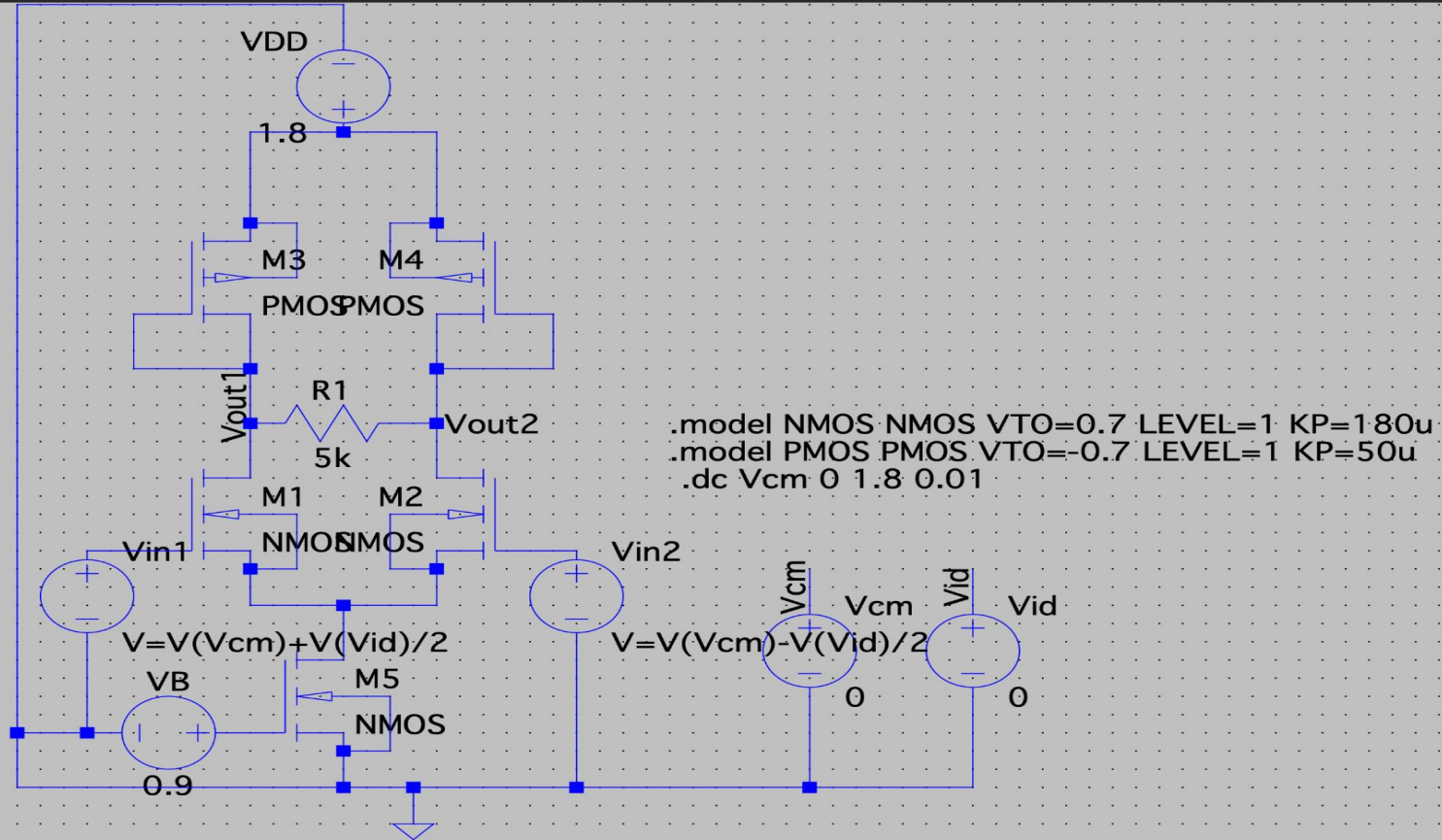
@ $I_s = \text{constant}$

$\therefore V_{out} = \text{constant}$

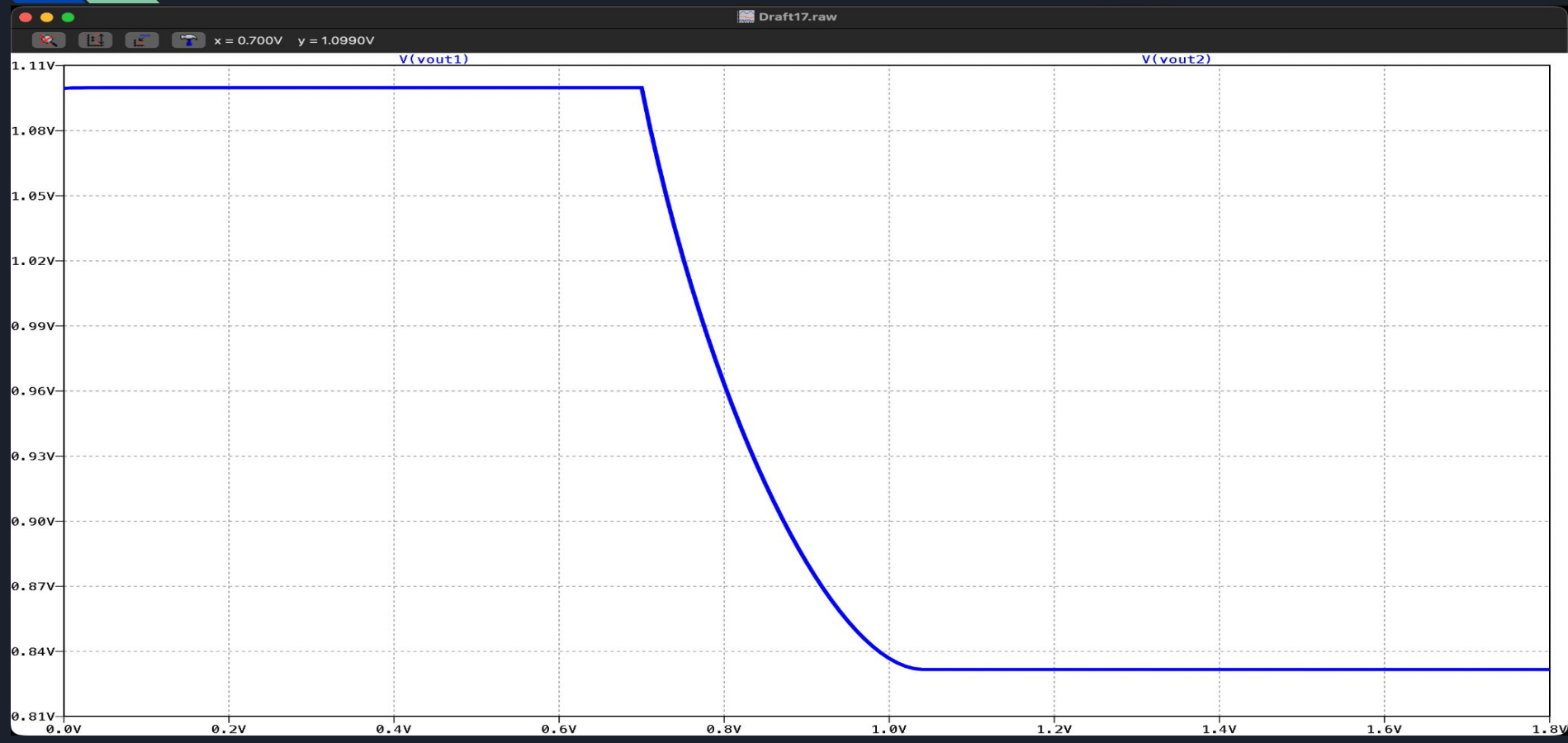


Plotted single-ended; ideal differential $V_{out} = 0$ V due to symmetry

LTSpice Schematic



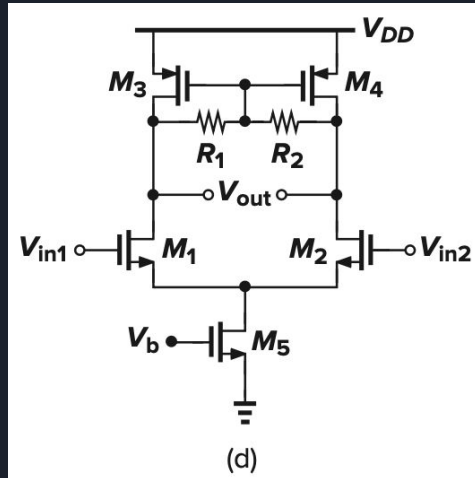
Simulation for Vout



Problem 4.16-d

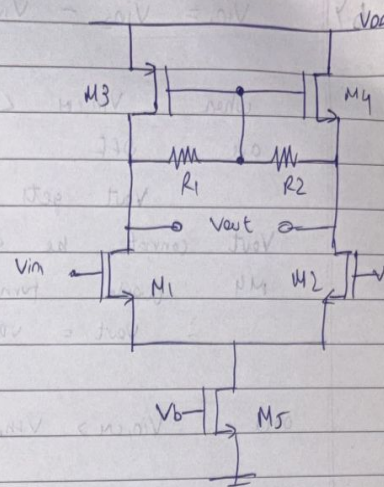
4.16. Assuming that all the circuits shown in Fig. 4.44 are symmetric, sketch V_{out} as (a) V_{in1} and V_{in2} vary differentially from zero to V_{DD} , and (b) V_{in1} and V_{in2} are equal and vary from zero to V_{DD} .

Diagram -



Handwritten Solution

Activity



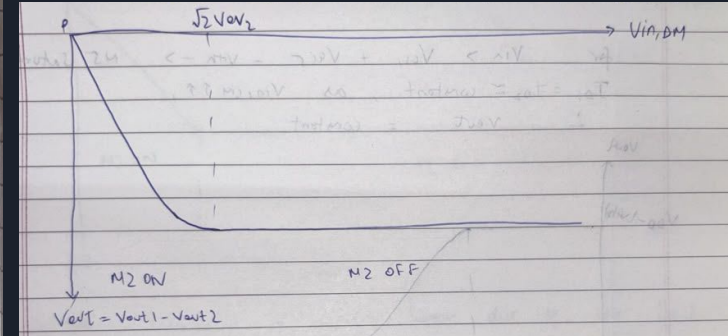
a) $V_{in1} - V_{in2} = V_{in,CM}$ varies from 0 to V_{DD}

When $V_{in1} - V_{in2} = 0$
 $I_{D1} = I_{D2} \therefore V_{out} = 0V$

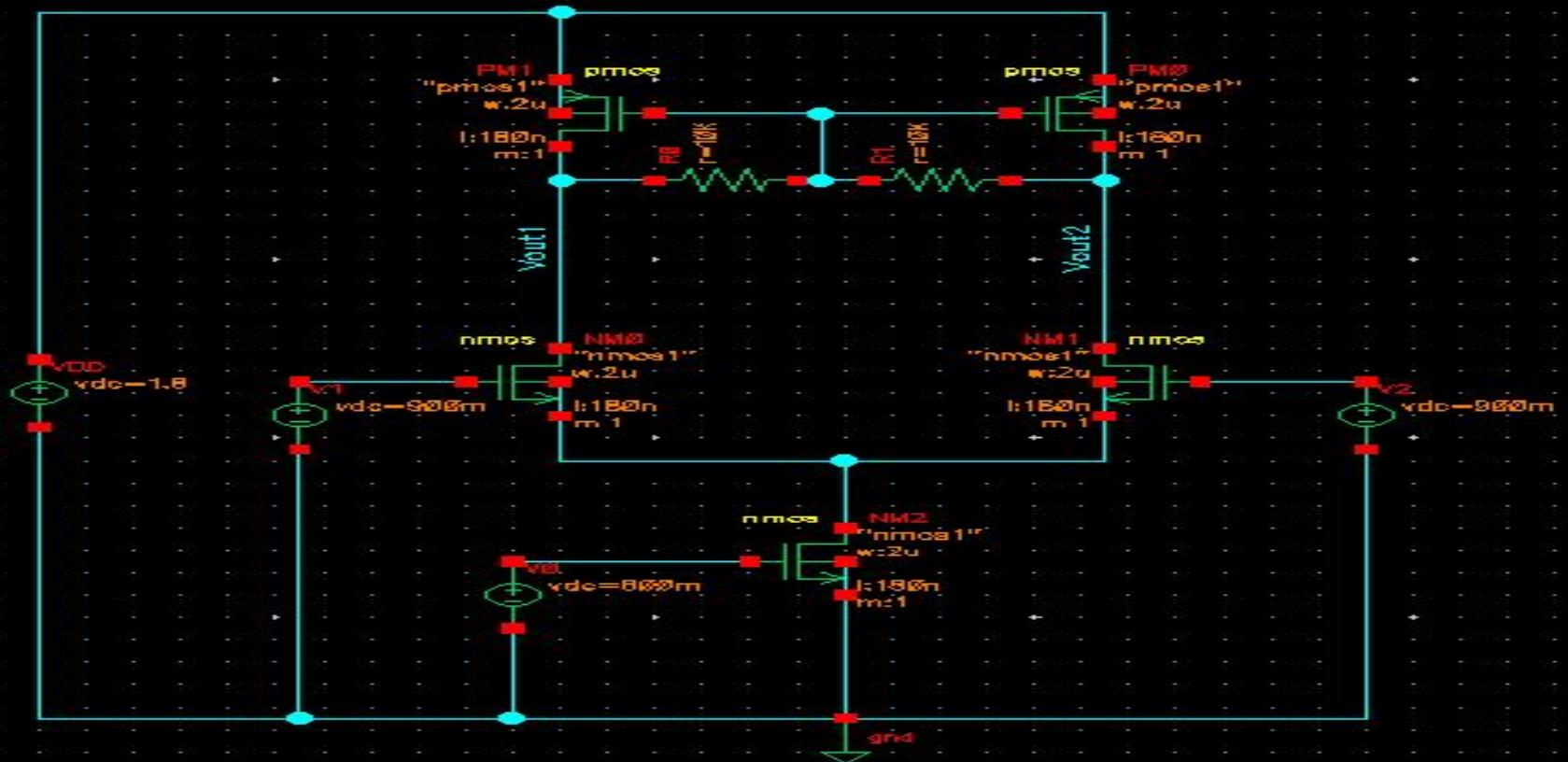
as $V_{in,CM}$ increases
 $I_{D1} \uparrow, I_{D2} \downarrow$

$\therefore V_{out} = V_{out1} - V_{out2}$
 V_{out1} decreases and V_{out2} increases

$\therefore V_{out} \downarrow \downarrow$, until M2 is ON
 and, M2 turns OFF
 V_{out} becomes constant, for $V_{in,CM} = \sqrt{2} V_{ov2}$



Cadence Virtuoso Schematic



Cadence ADE-L DC Simulation for Vout1 & Vout2

DC Response

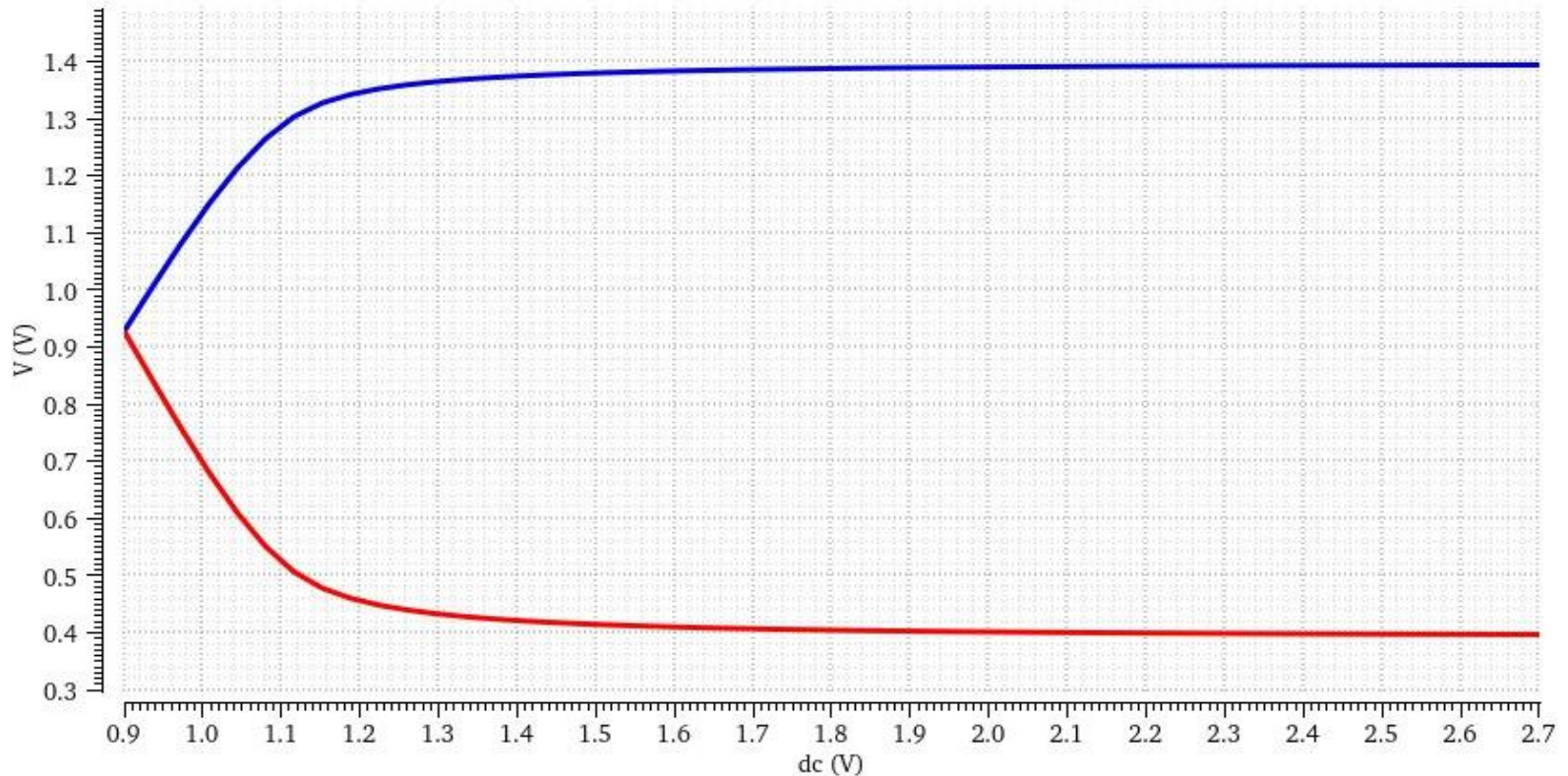
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Name Vis

/Vout2



/Vout1



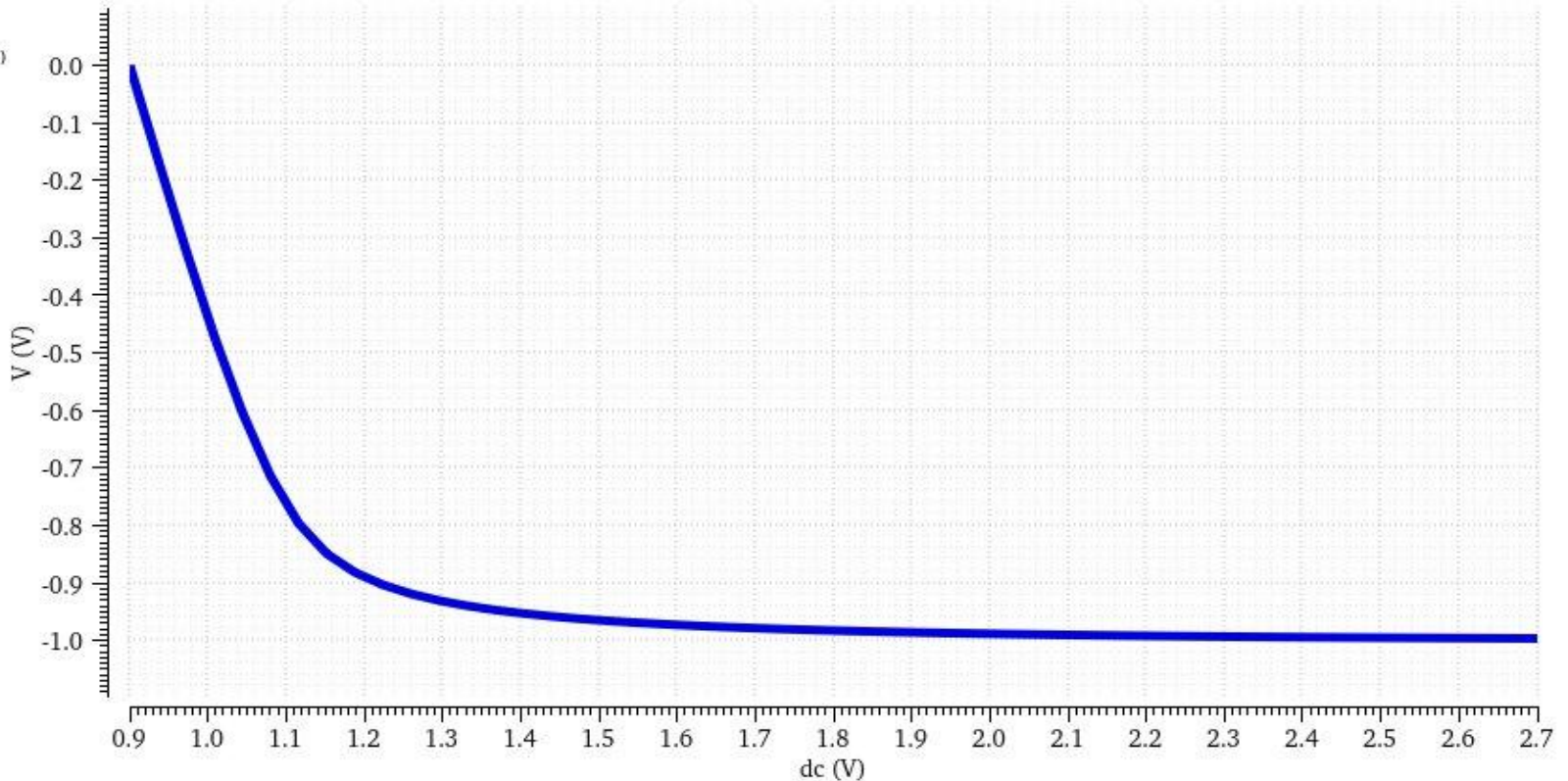
Cadence ADE-L DC Simulation for Vout

DC Response

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Name

■ ...-v("/Vout2" ?result "dc")



Handwritten Solution

6.7 $V_{in1} = V_{in2} = V_{in,CM}$ varies from 0 to V_{DD}

when $V_{in,CM} < V_{thn}$, M_1 and M_2 are OFF

V_{out} gets pulled UP, at 0

V_{out} cannot be equal to V_{DD} as then M_3 and M_4 would turn off

$\therefore V_{out} = V_{DD} - |V_{thp}|$

once $V_{in,CM} > V_{thn}$, current starts flowing

When $V_{in,CM}$ is ~~small~~ small, M_5 will be in triode

$V_{in} < V_{GS} + V_{GS} - V_{thn} \rightarrow M_5$ triode

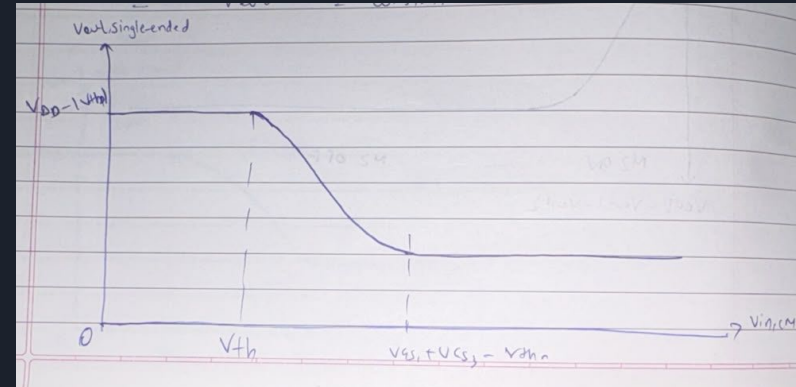
as $V_{in,CM} \uparrow$, I_{D1} and $I_{D2} \uparrow$, $\therefore$

V_{out} decreases

for $V_{in} > V_{GS} + V_{GS} - V_{thn} \rightarrow M_5$ Saturation

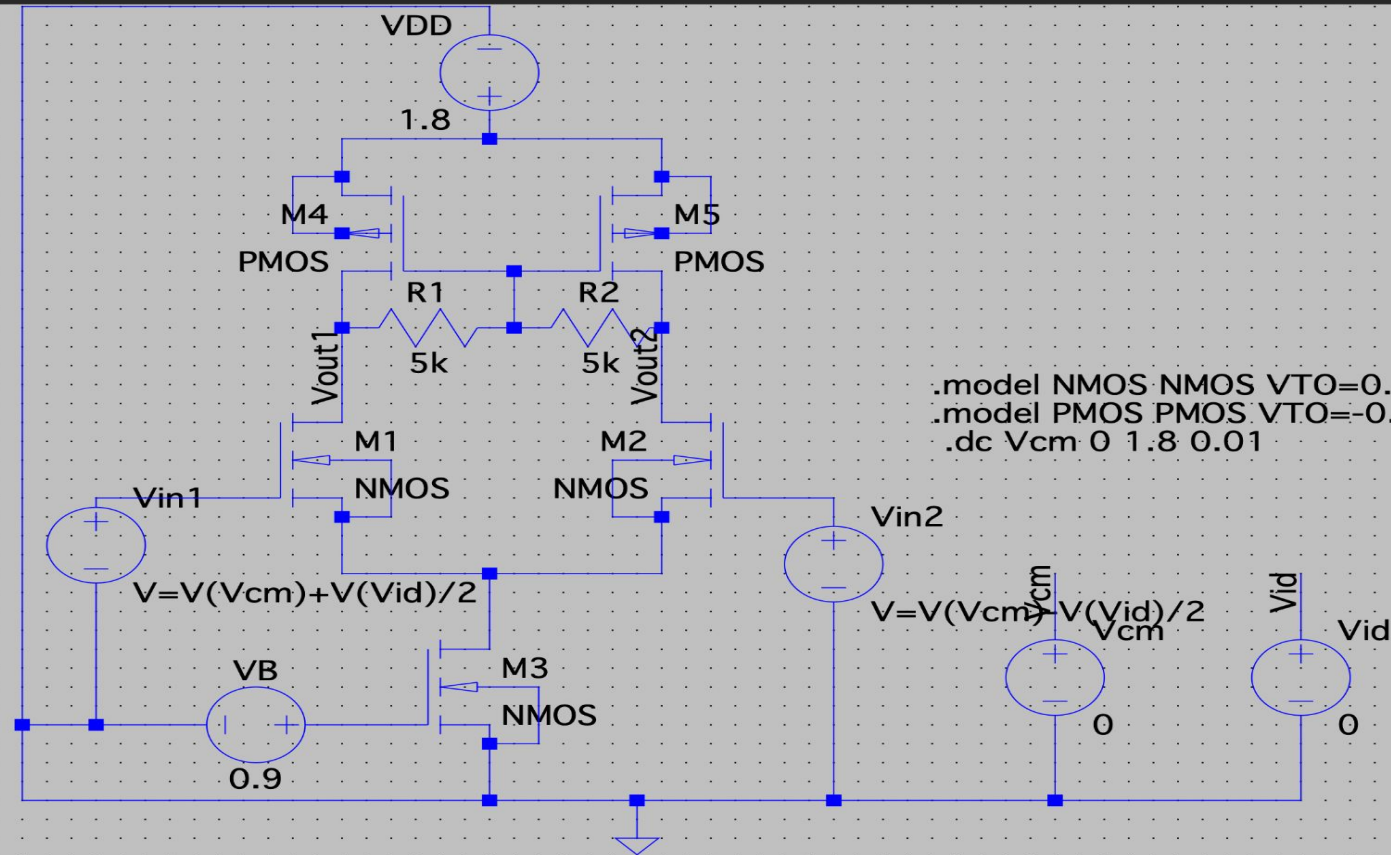
$I_{D1} = I_{D2} \approx \text{constant}$, as $V_{in,CM} \uparrow$,

$\therefore V_{out} = \text{constant}$



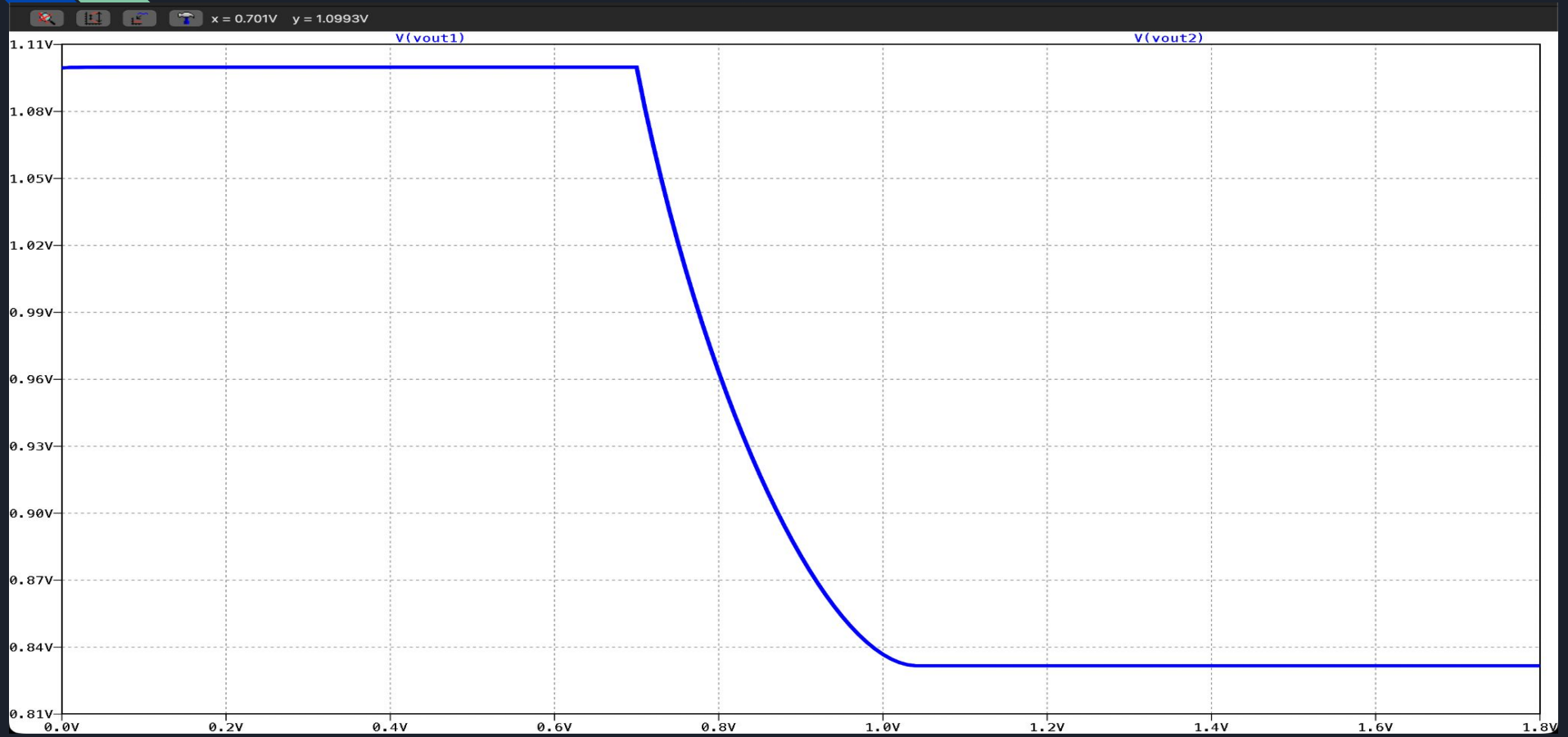
Plotted single-ended; ideal differential $V_{out} = 0$ V due to symmetry

LTSpice Schematic



```
.model NMOS NMOS VTO=0.7 LEVEL=1 KP=1.80u  
.model PMOS PMOS VTO=-0.7 LEVEL=1 KP=50u  
.dc Vcm 0 1.8 0.01
```

Simulation for Vout





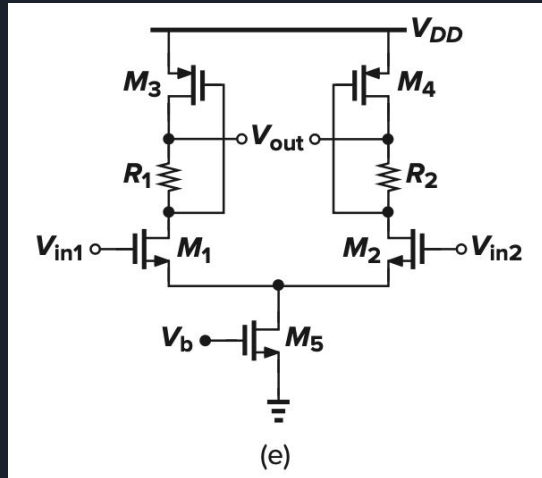
Points to be noted

- Under differential input conditions, V_{out1} and V_{out2} change symmetrically, causing the center tap voltage V_G of M3 and M4 to remain constant (AC virtual ground).
- Hence, the resistive network does not provide feedback for differential signals and acts as a passive differential load.
- For common-mode inputs, both output voltages change in the same direction, causing V_G to vary and creating negative feedback through the PMOS loads.
- This feedback adjusts the PMOS current and stabilizes the output voltages.
- When $V_{in,cm} < V_{thn}$, M1 and M2 turn OFF, and the outputs settle near $V_{DD} - |V_{thp}|$, where the PMOS devices reach the zero-current condition.

Problem 4.16-e

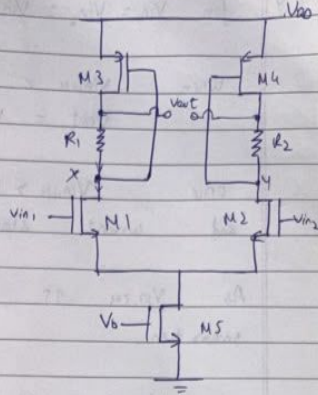
4.16. Assuming that all the circuits shown in Fig. 4.44 are symmetric, sketch V_{out} as (a) V_{in1} and V_{in2} vary differentially from zero to V_{DD} , and (b) V_{in1} and V_{in2} are equal and vary from zero to V_{DD} .

Diagram -



Handwritten Solution

Q.4 $V_{in} - V_{in2} = V_{in,ON}$ is varied from 0 to V_{DD}

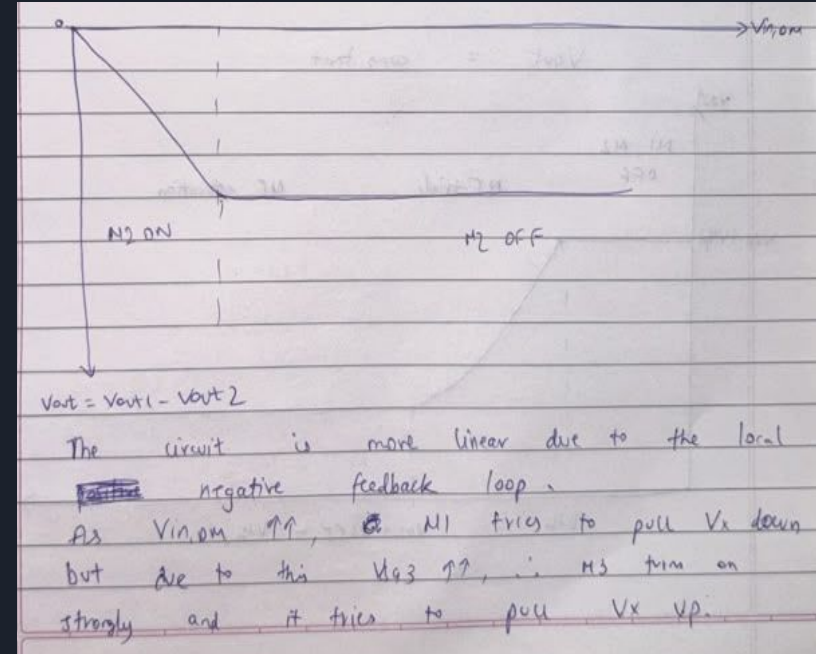


As $V_{in,ON}$ increases

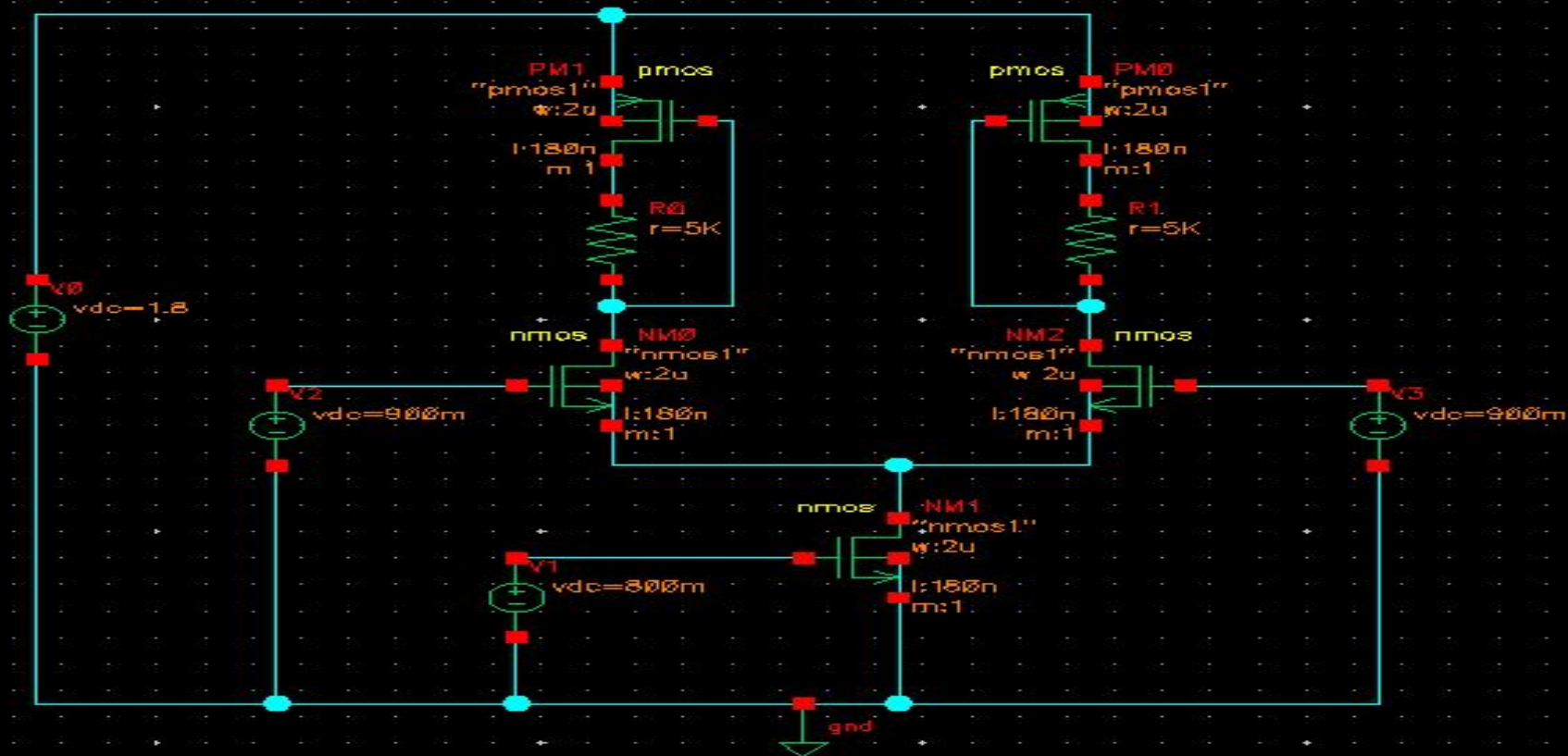
$I_{D1} \uparrow, I_{D2} \downarrow \therefore V_{out1} \downarrow$
and V_{out2} increases

$V_{out} = V_{out1} - V_{out2}$
 $\therefore V_{out}$ decreases

This is only valid till $M2$ is ON, once $M2$ turns off, entire current flows through $M1$, V_{out} becomes constant



Cadence Virtuoso Schematic



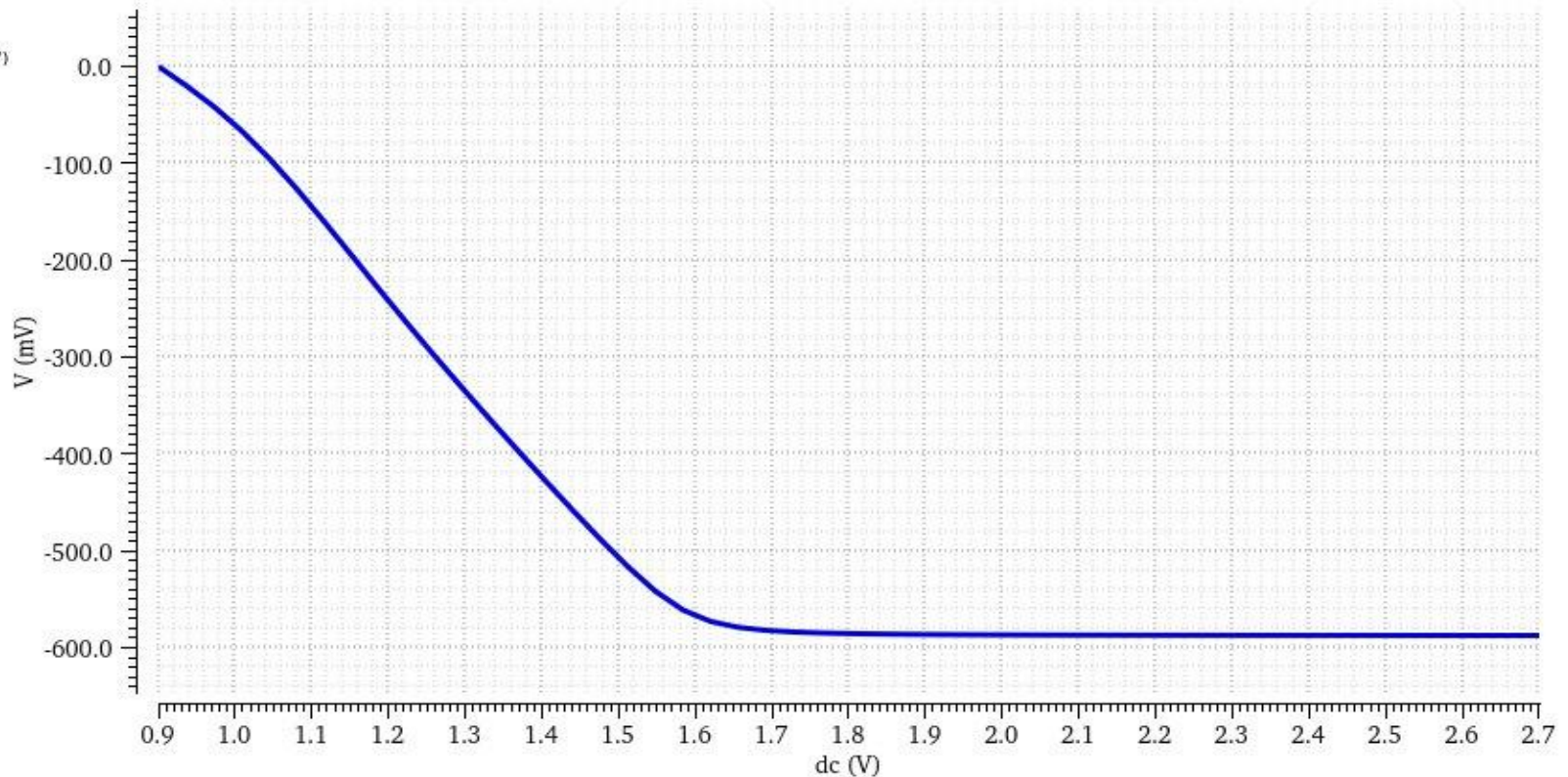
Cadence ADE-L DC Simulation for Vout

DC Response

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Name

...-v("/Vout2" ?result "dc")



Handwritten Solution

b. y for $V_{in1} = V_{in2} = V_{in,cm}$, varied from 0 to V_{DD}

When $V_{in,cm} < V_{thn}$, $M1$ and $M2$ are off
 $V_{out} = V_{DD} - |V_{thp}|$

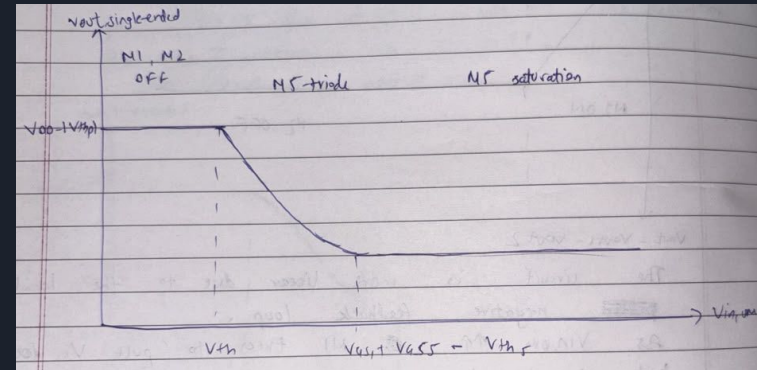
Once $V_{in,cm} > V_{thn}$, $M1$ and $M2$ turn ON and current starts to flow through them

As $V_{in,cm} \uparrow$, $V_{out} \downarrow$ till $M5$ enters saturation

$V_{in,cm} = V_{GS1} + V_{GS5} - V_{thf} \rightarrow$ boundary condition for $M5$

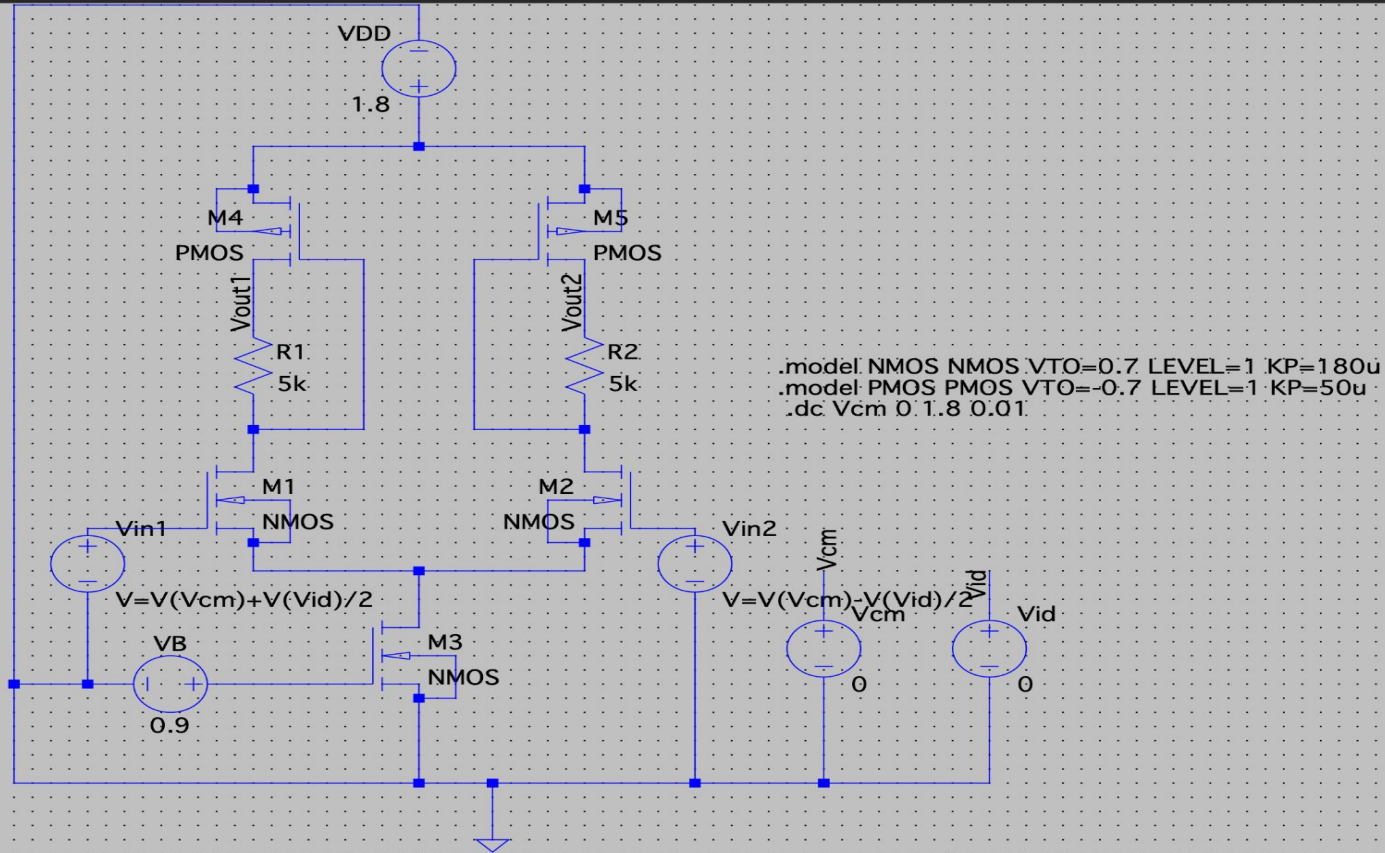
Once $M5$ enters saturation region, current becomes constant, thus further increase in $V_{in,cm}$ does not change V_{out}

$V_{out} = \text{constant}$

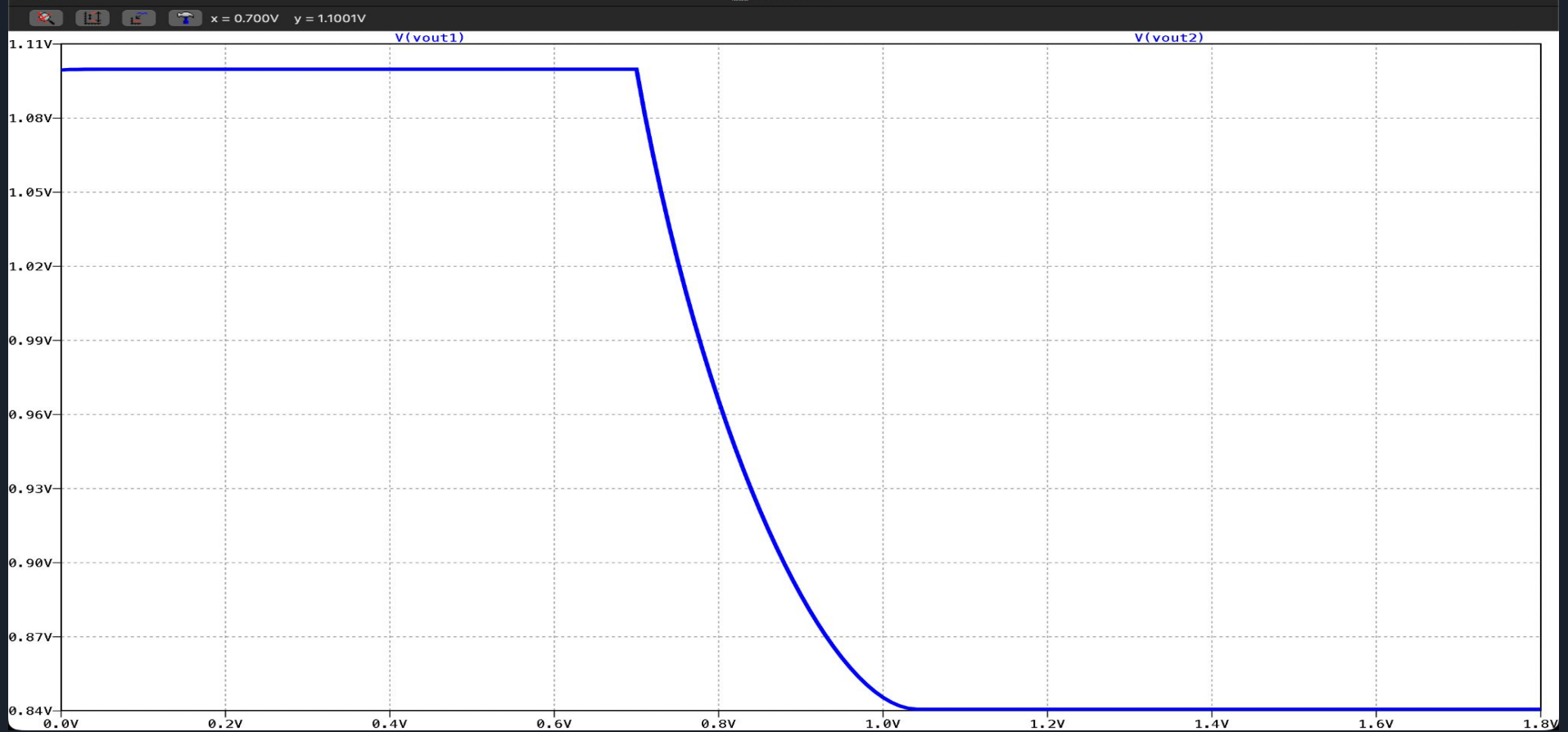


Plotted single-ended; ideal differential $V_{out}=0$ V due to symmetry

LTSpice Schematic



Simulation for Vout





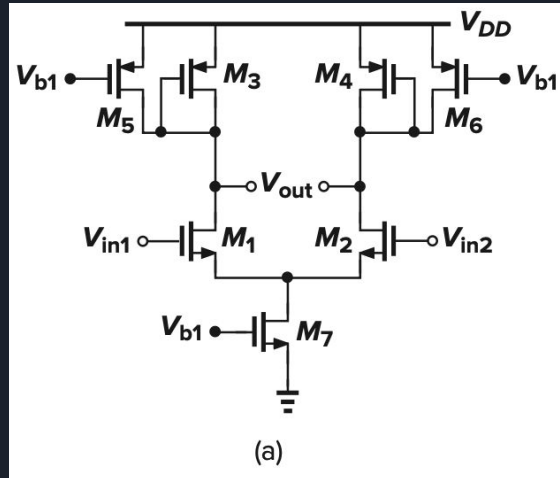
Points to be noted

- The circuit uses an NMOS differential pair with PMOS loads, where R1 and R2 introduce local series feedback by connecting each PMOS gate to the drain node of the corresponding NMOS branch.
- Under differential input, an increase in one branch current creates a voltage drop across the corresponding resistor, changing the PMOS gate voltage and increasing PMOS current to oppose the output variation.
- This local negative feedback improves output stability and linearity but reduces the differential voltage gain.
- Under common-mode input, both branches experience identical feedback action, maintaining symmetry and stabilizing the output response.
- When $V_{in,cm} < V_{thn}$, M1 and M2 turn OFF, causing no current through R1 and R2; the outputs settle near the PMOS cutoff boundary, approximately below VDD.

Problem 4.17-a

4.17. Assuming that all the circuits shown in Fig. 4.45 are symmetric, sketch V_{out} as (a) V_{in1} and V_{in2} vary differentially from zero to V_{DD} , and (b) V_{in1} and V_{in2} are equal and vary from zero to V_{DD} .

Diagram -



Handwritten Solution

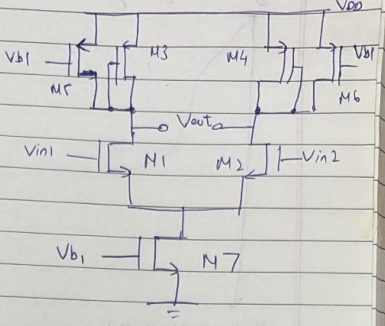
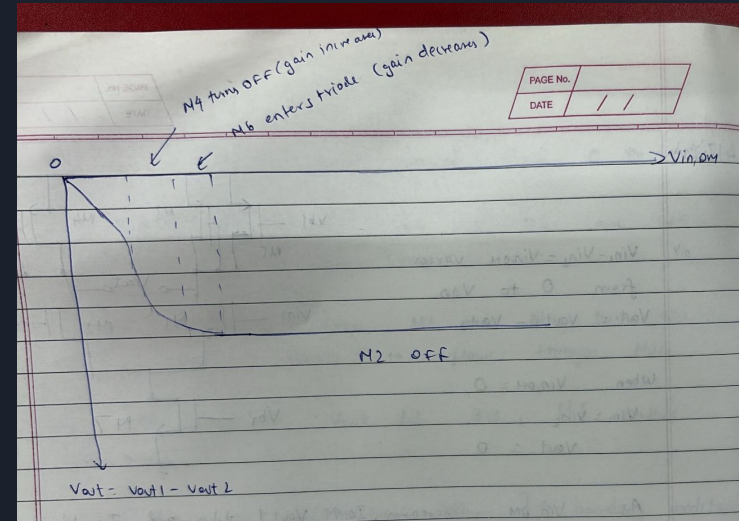
4.17.2014

a) $V_{in1} - V_{in2} = V_{in,CM}$ varies from 0 to V_{DD}
 $V_{out} = V_{out1} - V_{out2}$

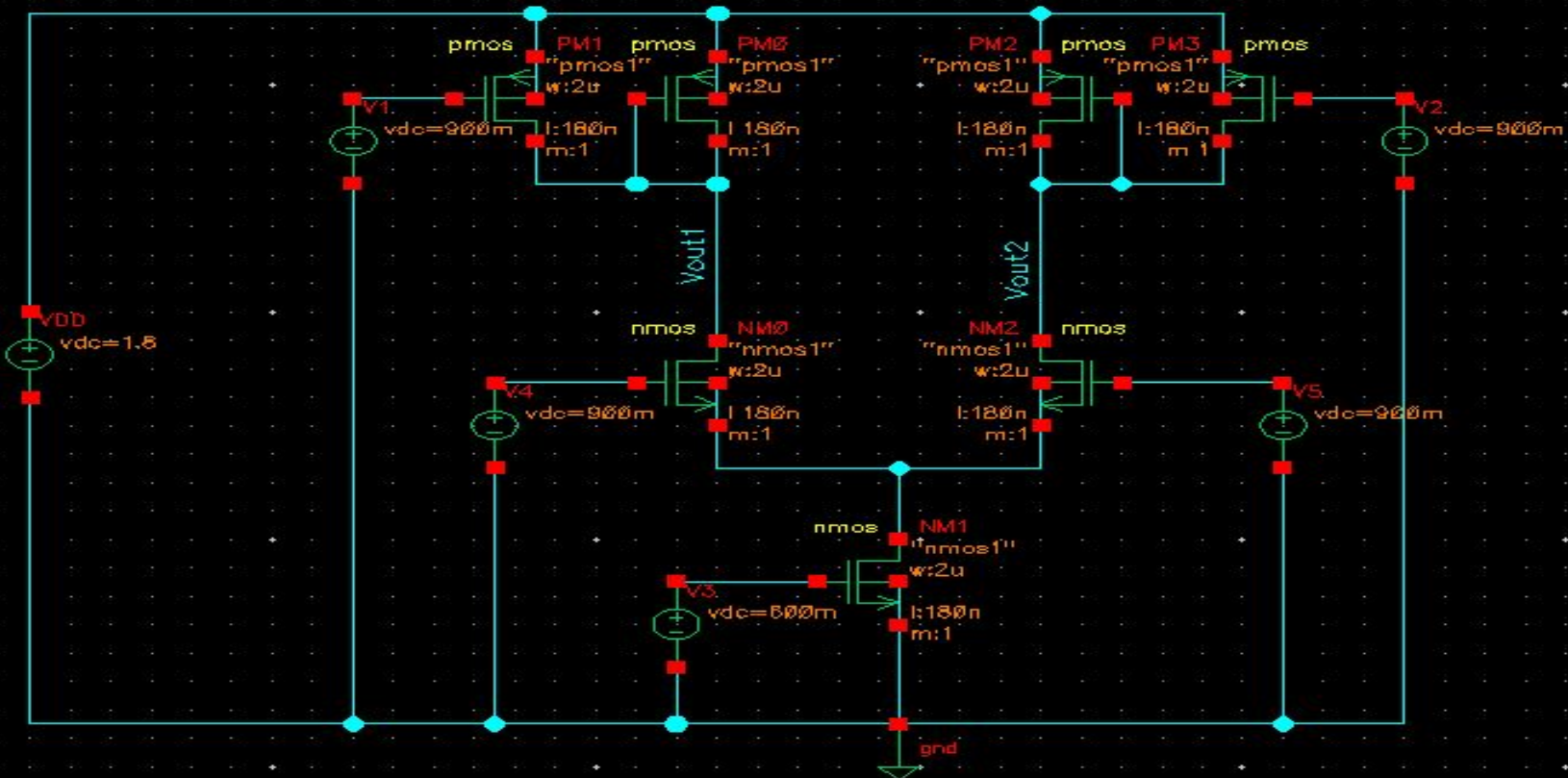
When $V_{in,CM} = 0$,
 $V_{in1} = V_{in2}$,
 $\therefore V_{out} = 0$

As $V_{in,CM}$ increases, $I_{D1} \uparrow$, $V_{out1} \downarrow$ and $I_{D2} \downarrow$, $V_{out2} \uparrow$
 $\therefore V_{out} \downarrow$

This is valid only till $M2$ is ON,
 once $M2$ turns OFF, V_{out} becomes constant
 → Also as $V_{in,CM}$ increases, V_{out2} increases, $\therefore$ after $V_{DD} - V_{out2} < |V_{thp1}|$, $M4$ turns off and gain increases sharply (R_{out} increases)
 → $M6$ will enter triode region when $V_{out2} > V_{DD} + |V_{thp1}|$
~~gain~~ R_{out} decreases and gain decreases

Cadence Virtuoso Schematic



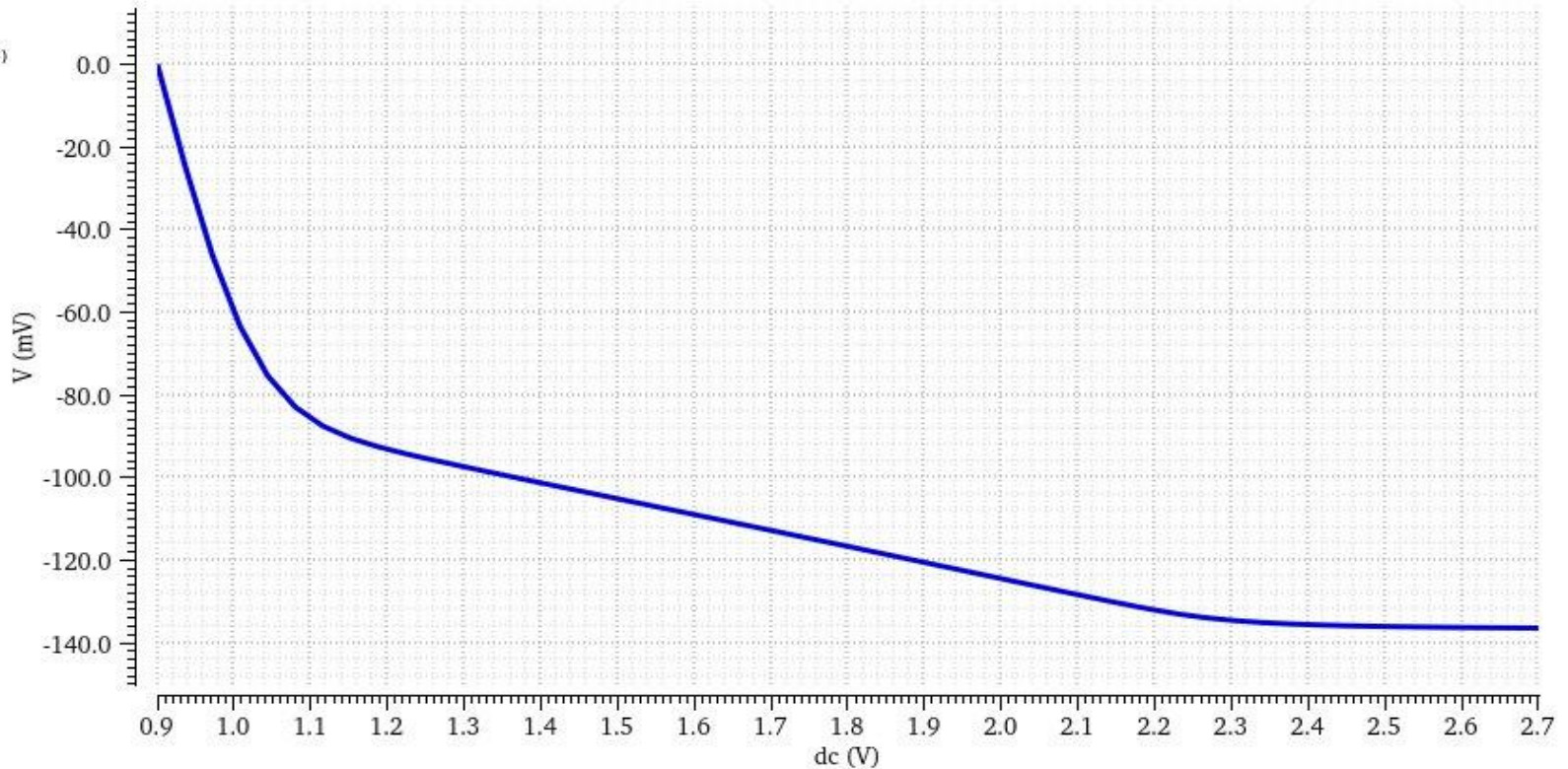
Cadence ADE-L DC Simulation for Vout

DC Response

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Name

...-v("/Vout2" ?result "dc")



Handwritten Solution

5.4 $V_{in_1} = V_{in_2} = V_{in,CM}$, varies from 0 to V_{DD}

When $V_{in,CM}$ is less than V_{thn} ,

M1 and M2 are OFF,

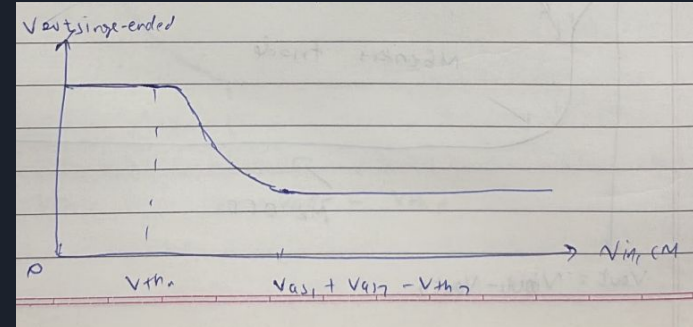
$$V_{out} = V_{DD}$$

Once $V_{in,CM} > V_{thn}$ and $V_{in,CM} < V_{gs_1} + V_{gs_2} - V_{thn}$,

M1 is in triode and as $V_{in,CM}$ increases, currents increase $\therefore V_{out} \downarrow$

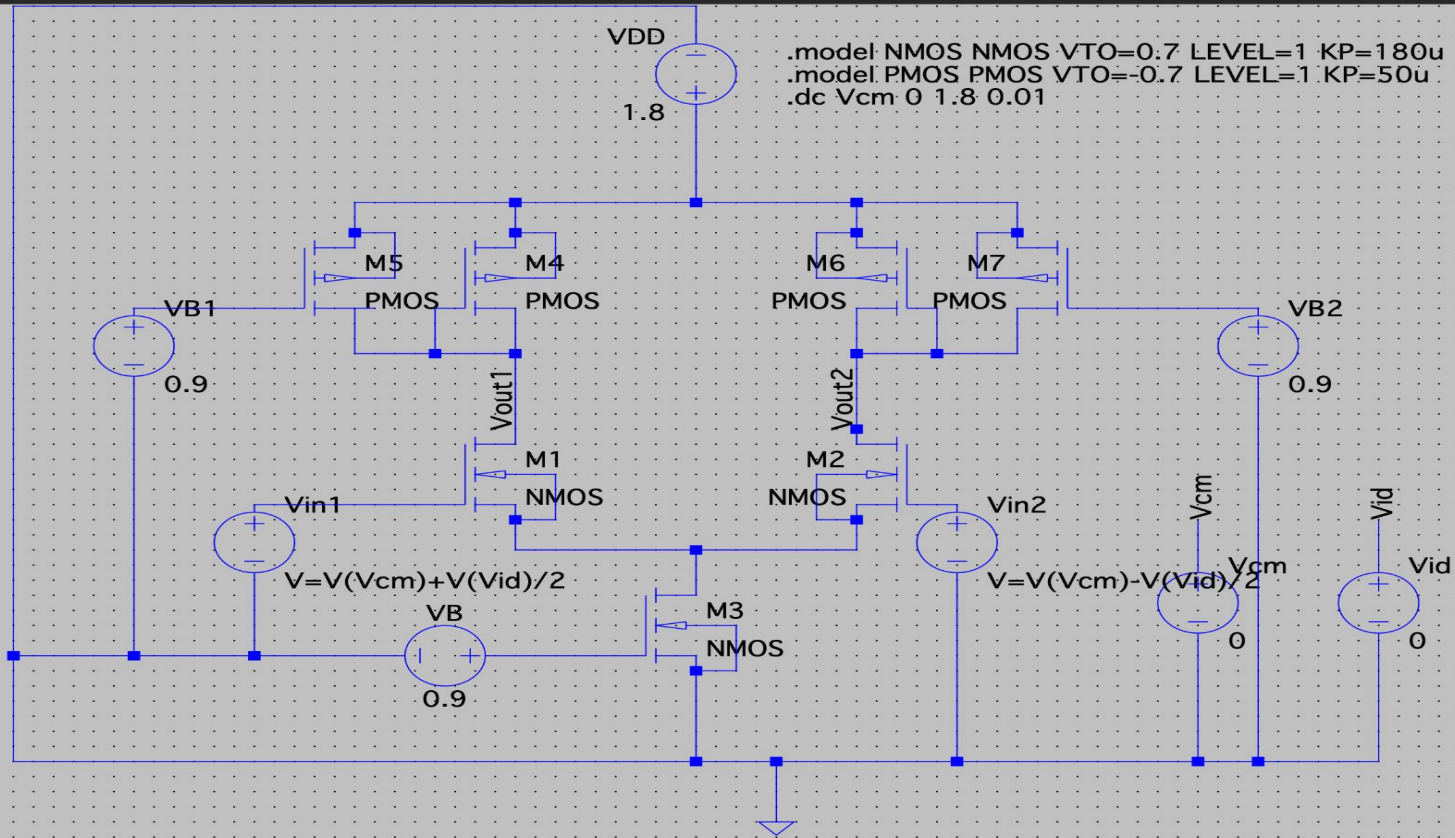
Once $V_{in,CM} > V_{gs_1} + V_{gs_2} - V_{thn}$, M1 enters saturation and current becomes constant

$$\therefore V_{out} = \text{constant}$$

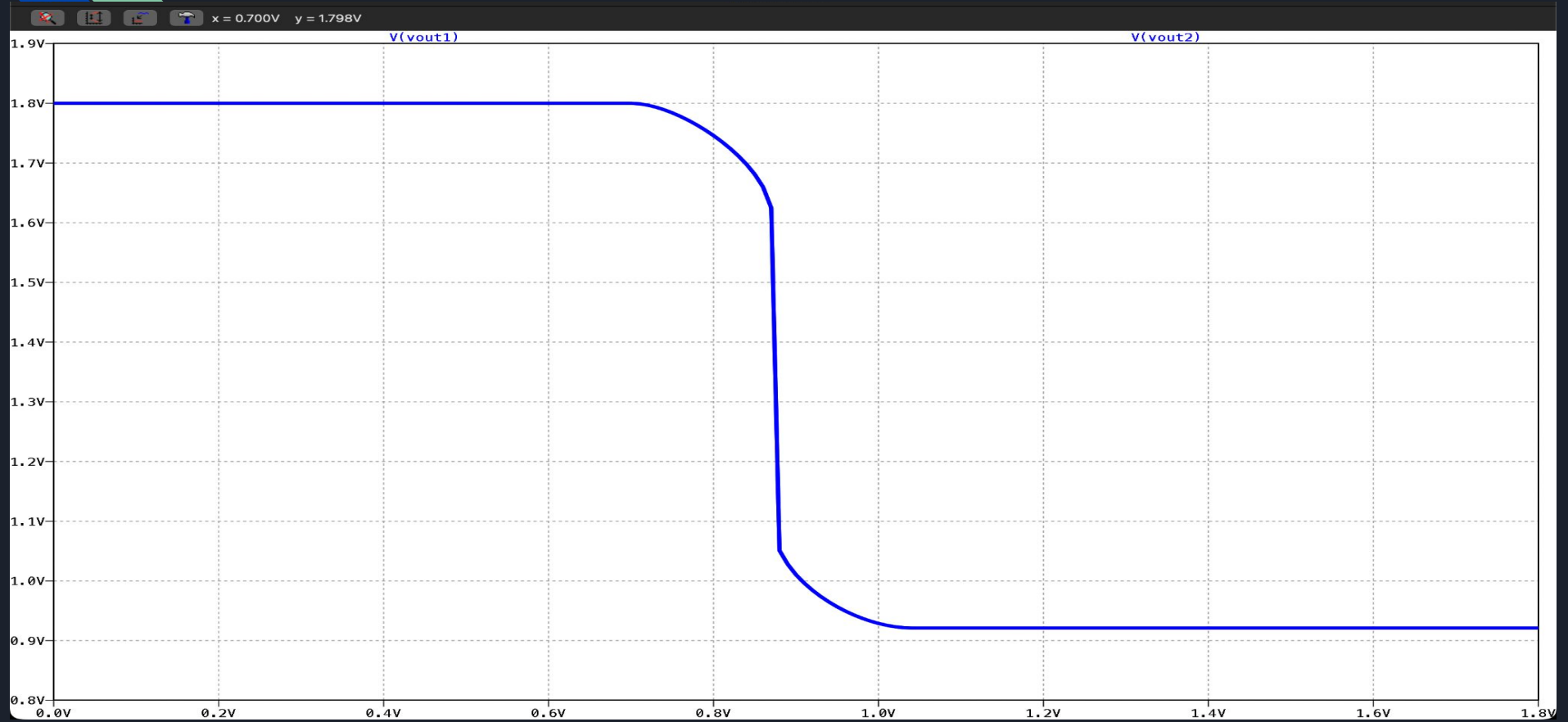


Plotted single-ended; ideal differential $V_{out} = 0$ V due to symmetry

LTSpice Schematic



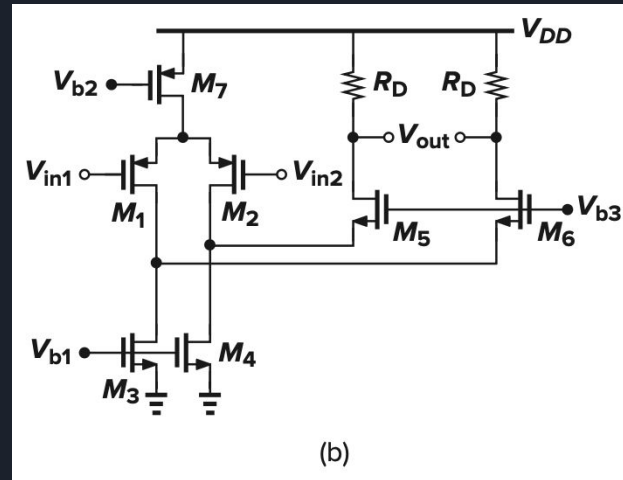
Simulation for Vout



Problem 4.17-b

4.17. Assuming that all the circuits shown in Fig. 4.45 are symmetric, sketch V_{out} as (a) V_{in1} and V_{in2} vary differentially from zero to V_{DD} , and (b) V_{in1} and V_{in2} are equal and vary from zero to V_{DD} .

Diagram -



4-17-4b

$\therefore$ for $V_{in,cm} = V_{in1} - V_{in2}$
varies from 0 to V_{DD}

When $V_{in1} \uparrow$ and $V_{in2} \downarrow$
[$V_{in,cm} \uparrow$]

The $V_{S4} \downarrow$, and $V_{S3} \uparrow$
 $I_{M1} \downarrow$ and $I_{M2} \uparrow$

Due to this V_x will decrease
and V_y will increase

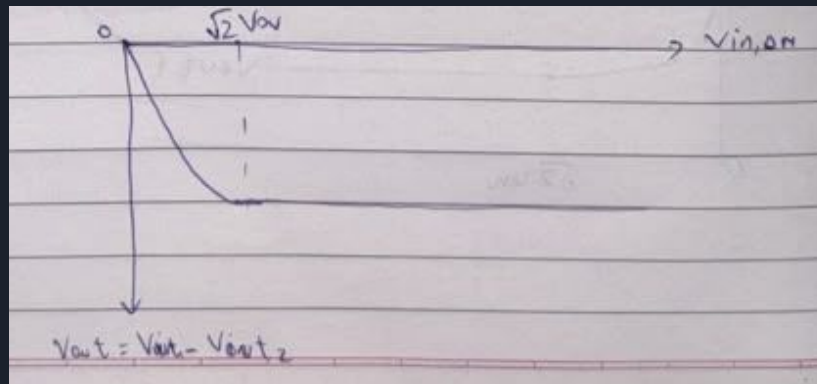
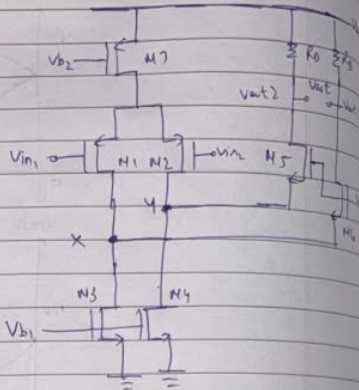
Due to decrease in V_x V_{S6} increases
Due to increase in V_y V_{S5} decreases

$\therefore$ $I_{M6} \uparrow$ and $I_{M7} \downarrow$

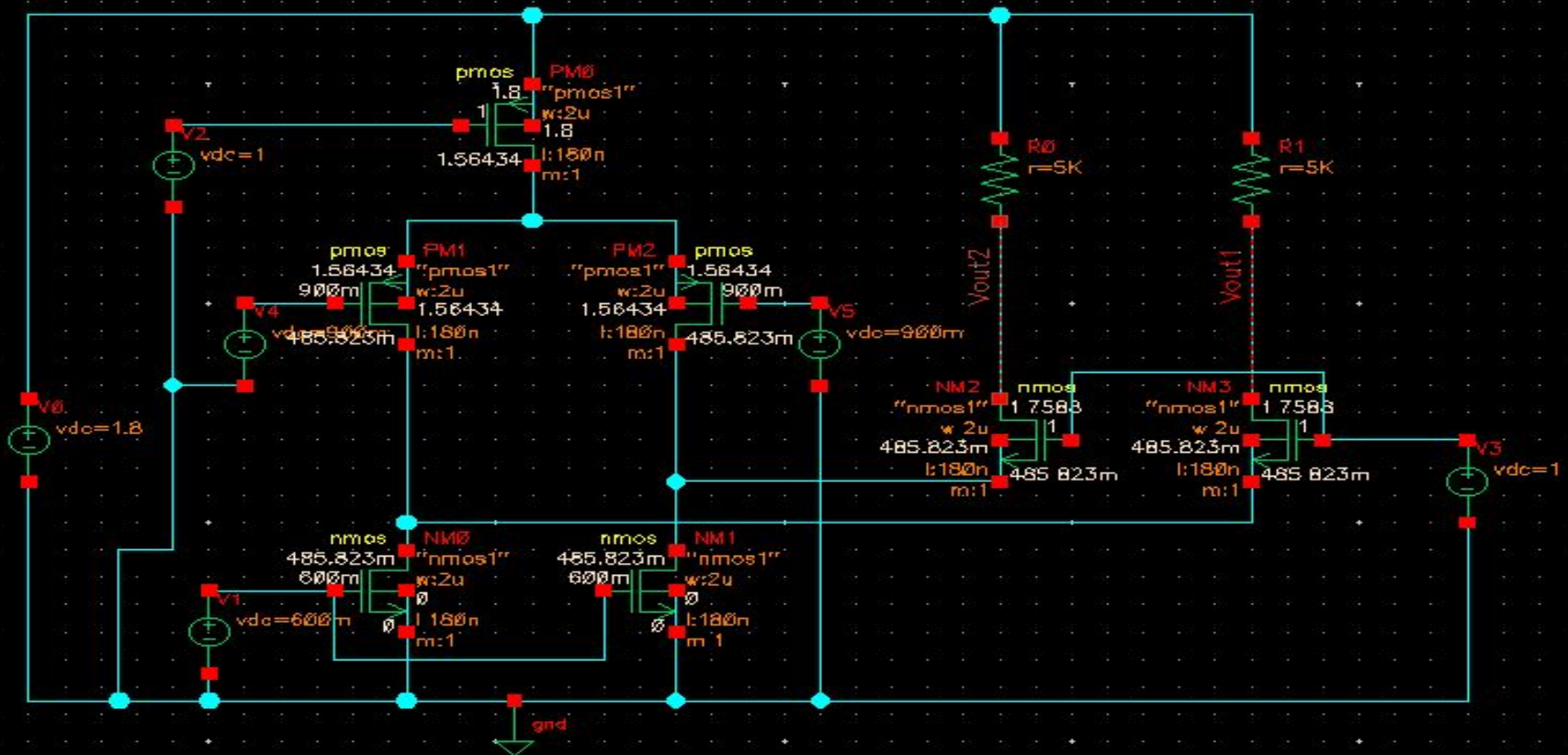
$\therefore$ $V_{out1} \downarrow$ decreases (drop across R_D increases)
 $V_{out2} \uparrow$ increases (drop across R_D decreases)

$\therefore$ $V_{out} = V_{out1} - V_{out2}$ decreases

This is valid only till $M1$ is ON
once $M1$ turns off, V_{out} becomes constant



Cadence Virtuoso Schematic



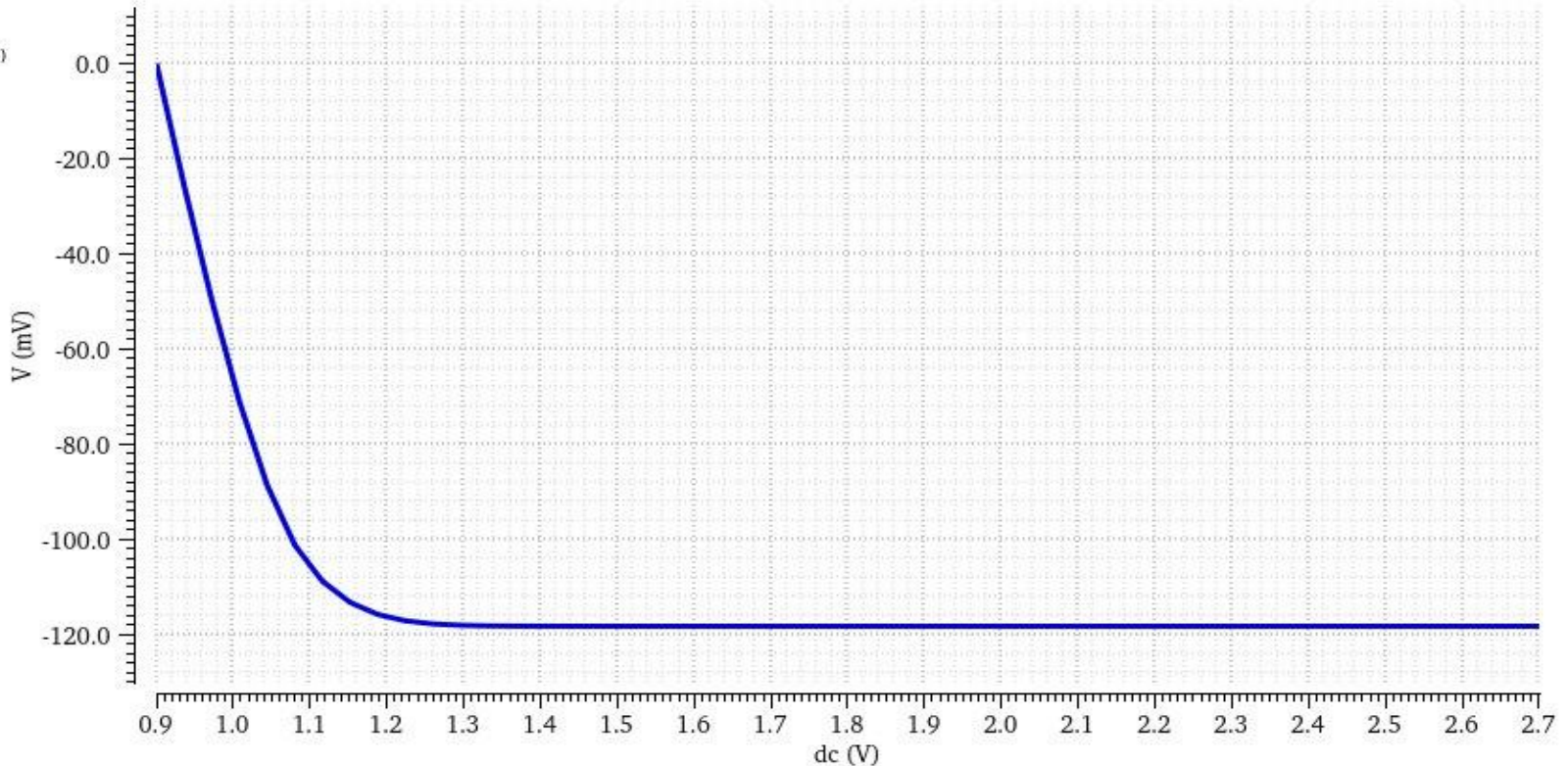
Cadence ADE-L DC Simulation for Vout

DC Response

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Name

...v("/Vout2" ?result "dc")

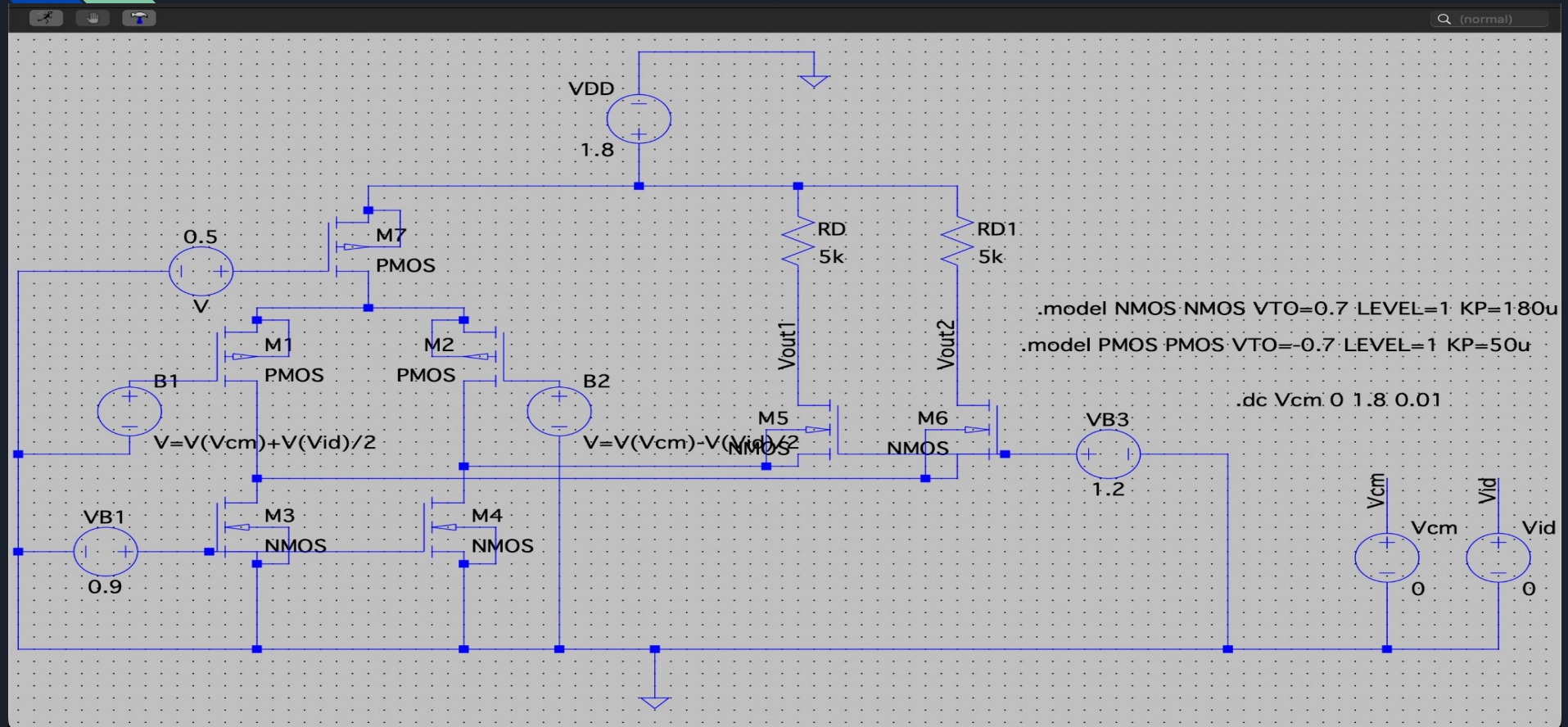




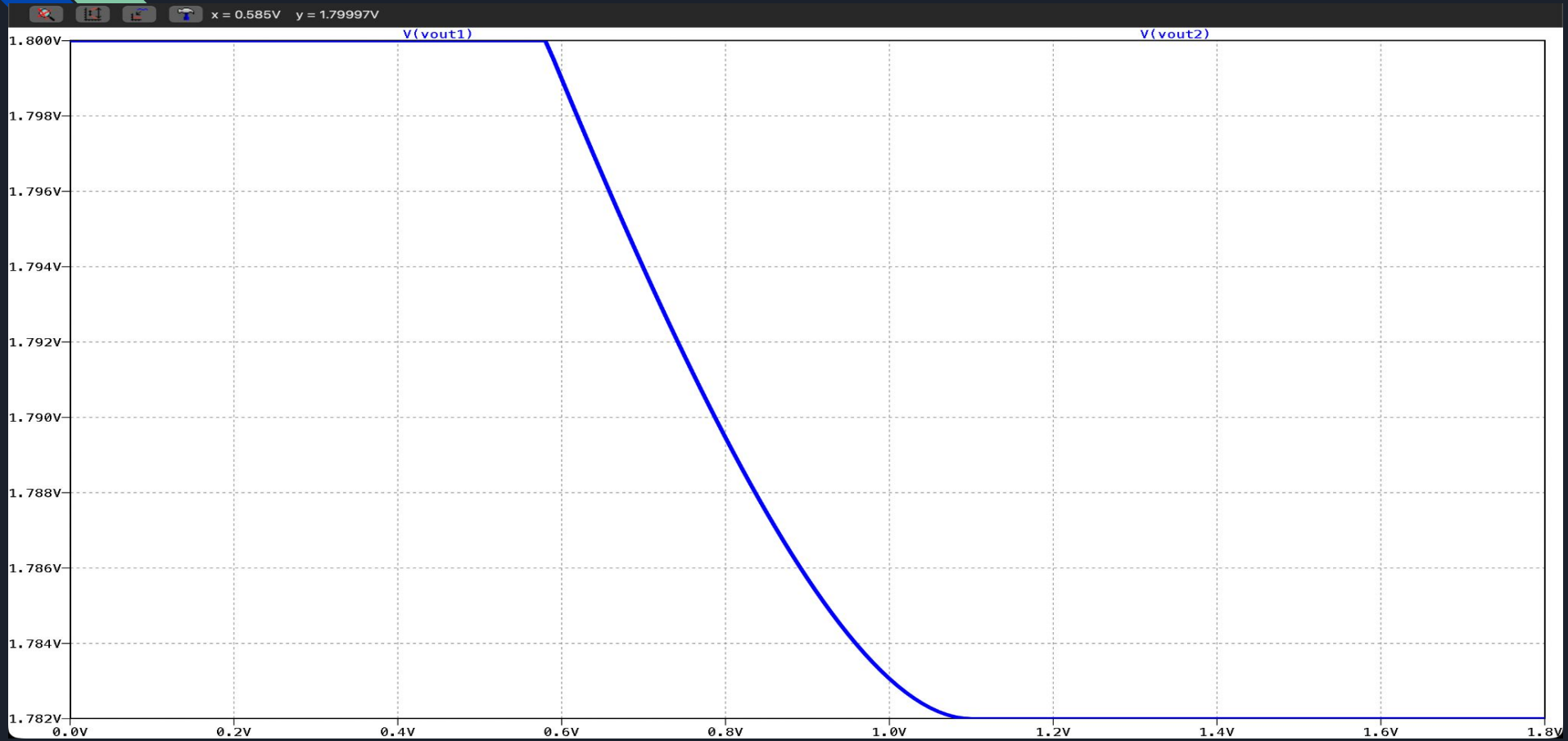
Solution for common mode characteristics

- When V_{in} is small, $M1$ and $M2$ are strongly turned ON hence the source nodes of $M5$ and $M6$ are at a high potential hence turning them OFF and making $V_{out} = V_{DD}$
- Once V_{in} rises enough so as to turn $M5$ and $M6$ ON, V_{out} starts to drop with further increase in V_{in} .
- Finally V_{out} becomes constant for large values of V_{in} as $M1$ and $M2$ turn OFF for large values of V_{in} , causing drop across R_D and R_{D1} to be constant even if V_{in} increases

LTSpice Schematic



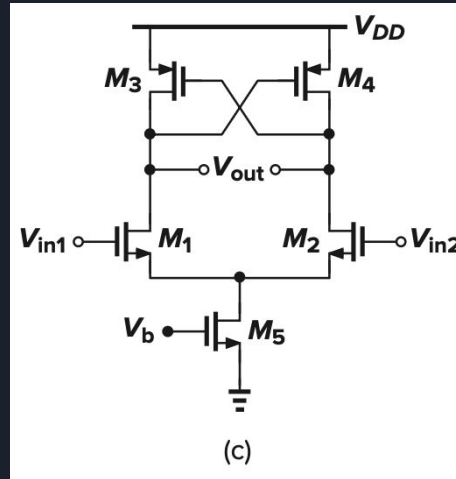
Simulation for Vout



Problem 4.17-c

4.17. Assuming that all the circuits shown in Fig. 4.45 are symmetric, sketch V_{out} as (a) V_{in1} and V_{in2} vary differentially from zero to V_{DD} , and (b) V_{in1} and V_{in2} are equal and vary from zero to V_{DD} .

Diagram -



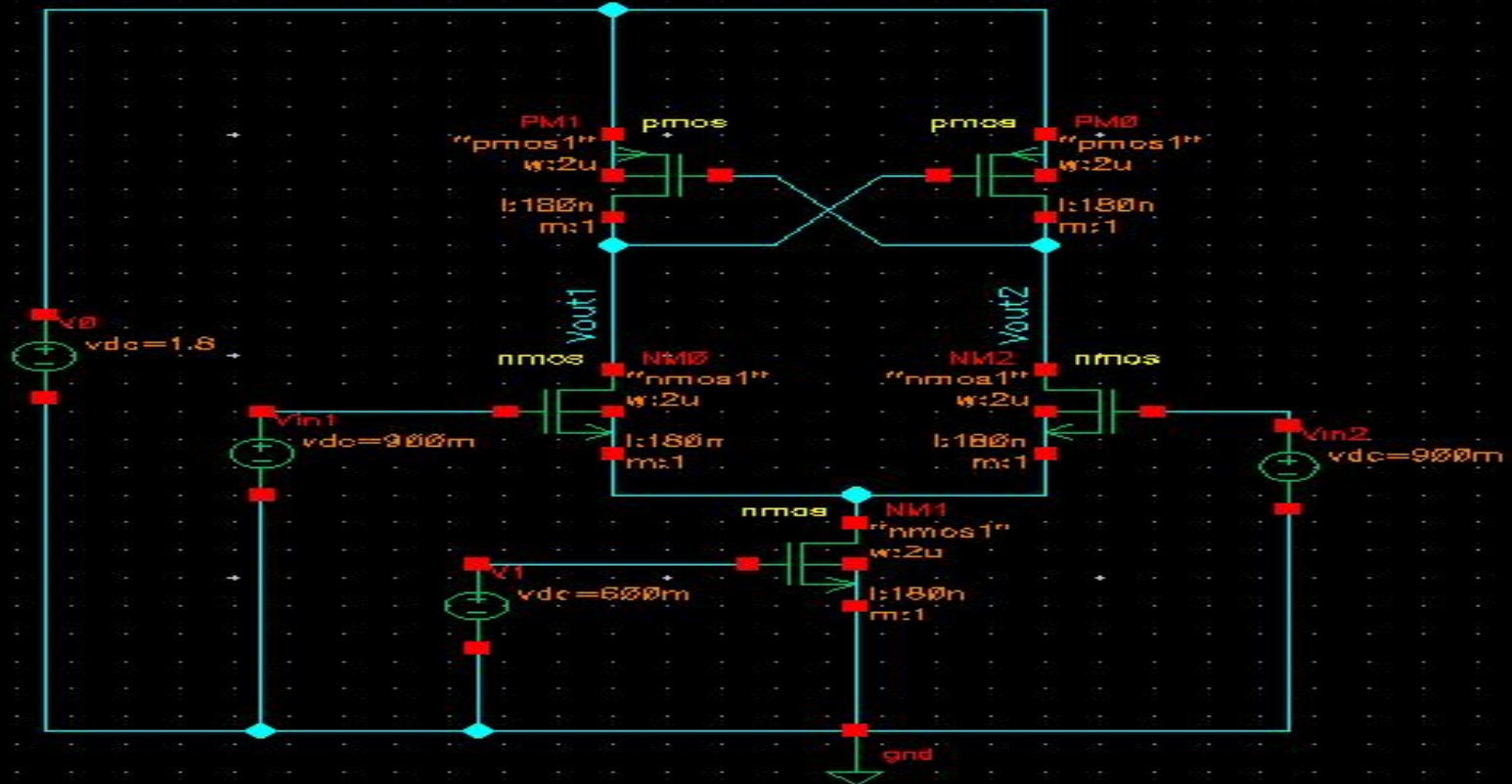


Solution

DC Graph Breakdown & Root Cause

- The **cross-coupled PMOS transistors introduce negative differential resistance**, resulting in a response different from a normal differential pair, where an increase in V_{in1} typically causes V_{out1} to decrease. Here, the positive-feedback action can instead reinforce changes at the output nodes i.e as V_{in1} increase V_{out1} also increases.
- As the differential input $V_{in1} - V_{in2}$ increases, **positive feedback becomes dominant**, driving the outputs toward opposite rails.
- Consequently, **V_{out1} decreases and saturates at a low voltage, while V_{out2} increases and approaches VDD.**

Cadence Virtuoso Schematic



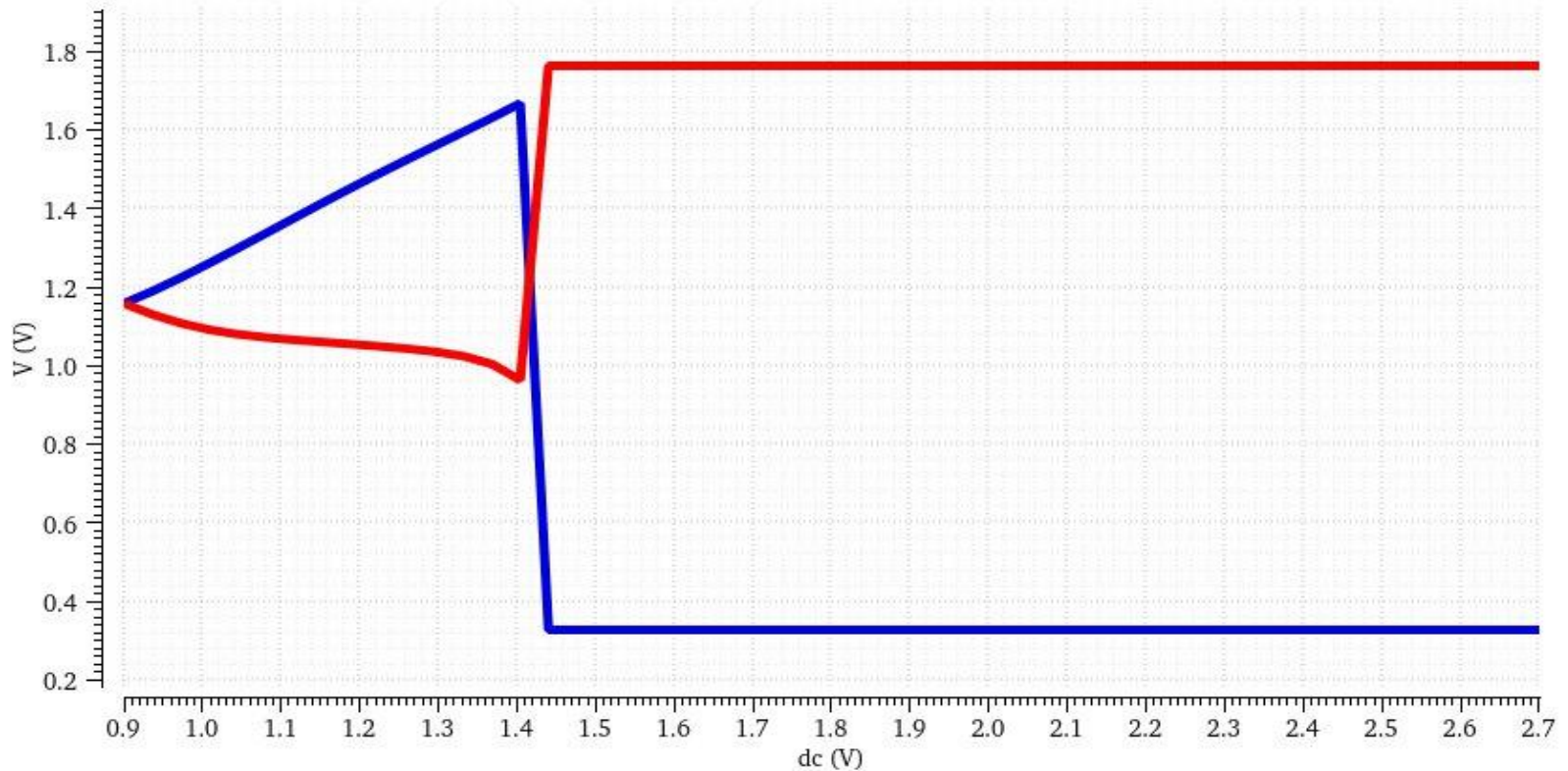
Cadence ADE-L DC Simulation for Vout

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DC Response

Name	Vis
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■ /Vout1	☒
■ /Vout2	☒



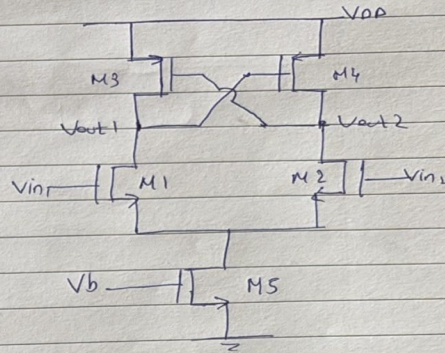
Handwritten Solution

for common mode operation

$$V_{in1} = V_{in2} = V_{cm}$$

when $V_{cm} < V_{th}$

M1 and M2 are OFF

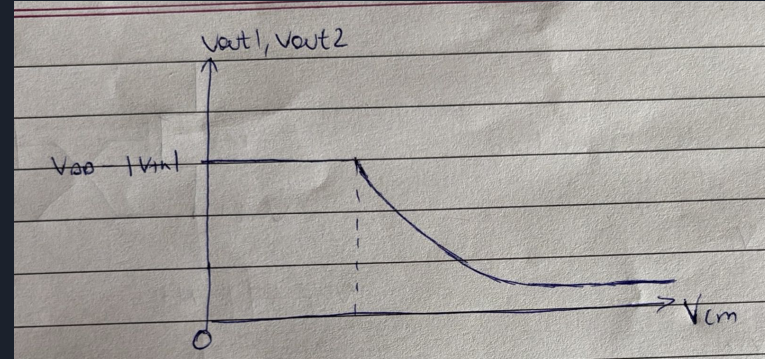


$$\therefore V_{out1} = V_{out2} = V_{DD} - 1V_{thp1} \\ = 1.8 - 0.7 = 1.1V$$

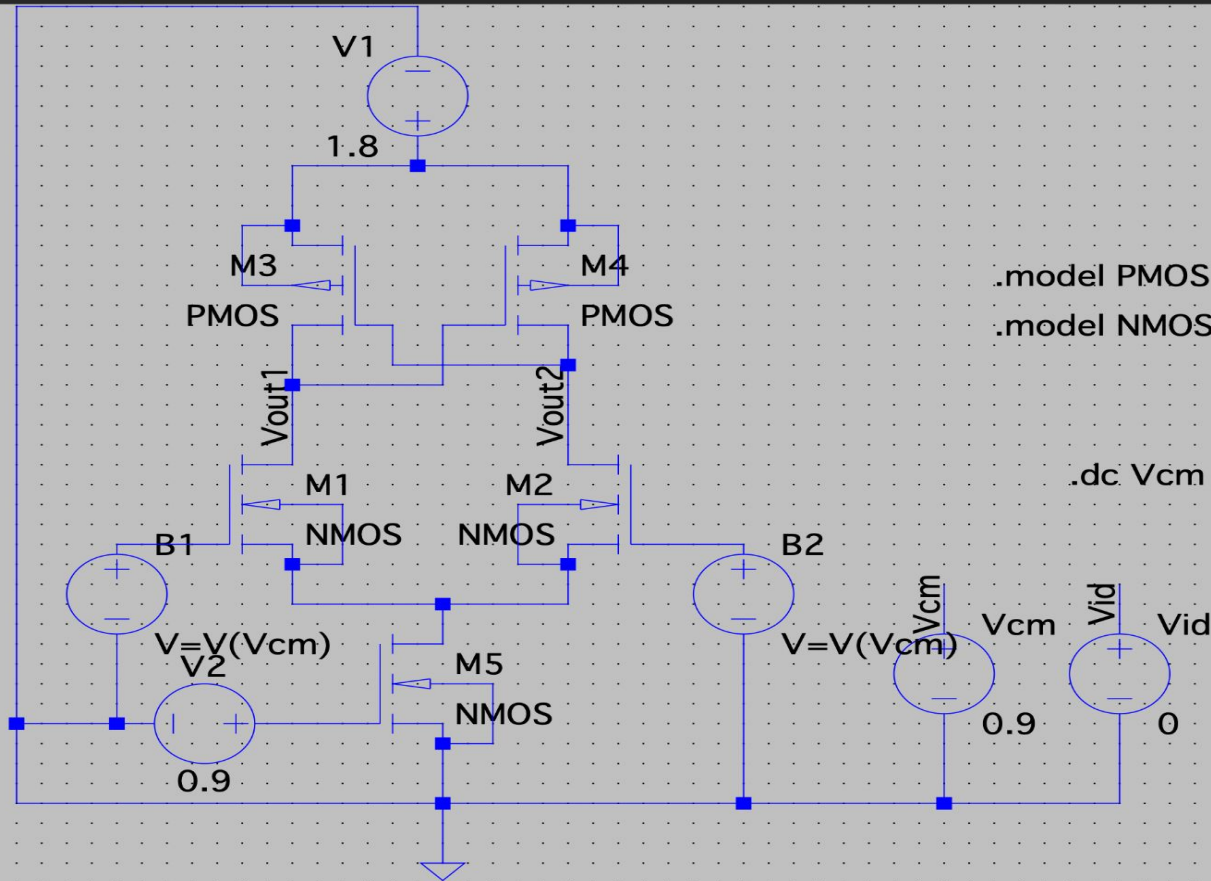
Once $V_{cm} > V_{th}$, M1 and M2 turn on in saturation.

$\therefore$ As $V_{cm} \uparrow \uparrow$, $V_{out1} = V_{out2}$ decrease

as V_{cm} becomes very large, M1 and M2 enter triode region and $V_{out1} = V_{out2}$ get pulled down and become constant.



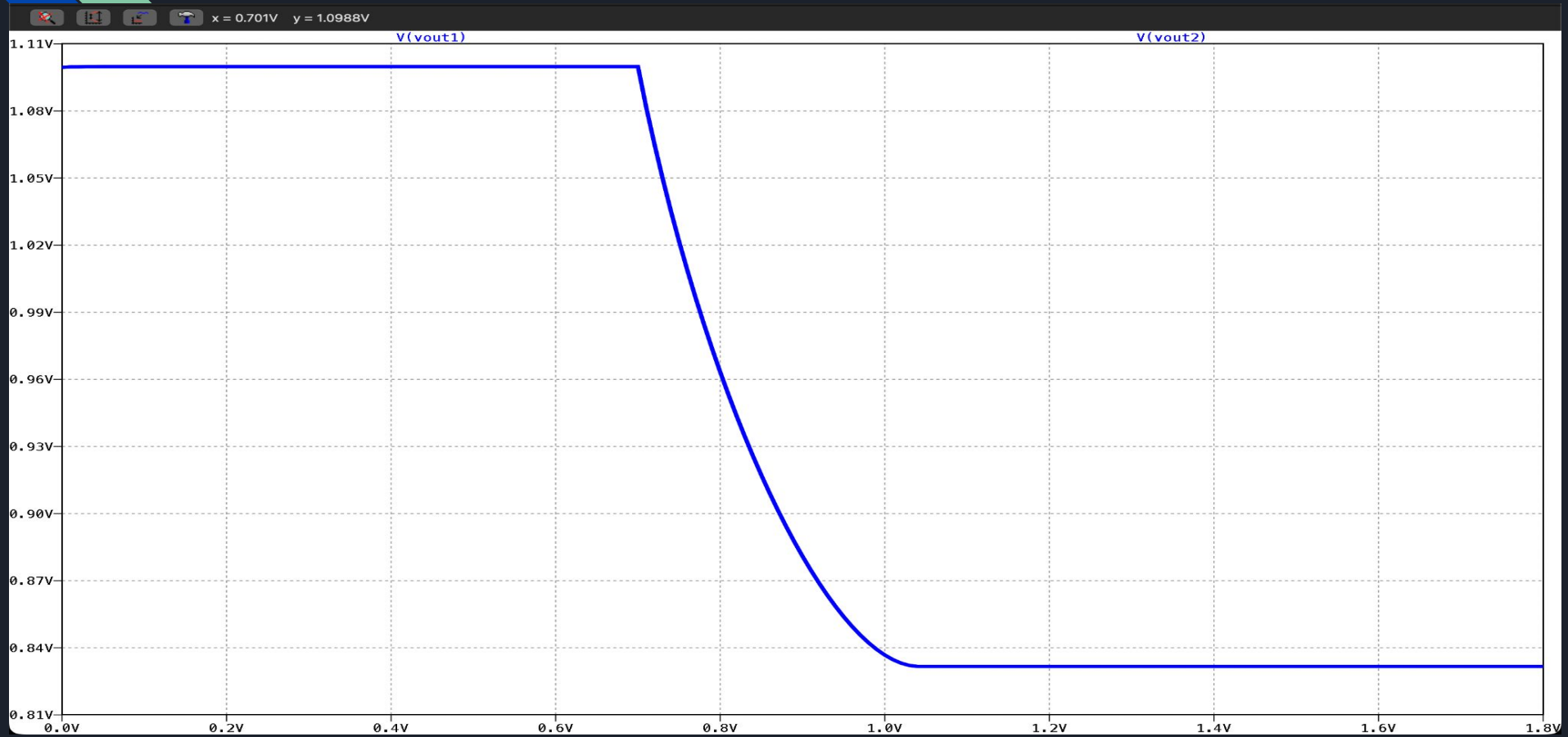
Plotted single-ended; ideal differential $V_{out} = 0V$ due to symmetry



```
.model PMOS PMOS VTO=-0.7 LEVEL=1 KP = 50u
.model NMOS NMOS VTO=0.7 LEVEL=1 KP = 180u
```

```
.dc Vcm 0 1.8 0.01
```

Simulation for Vout



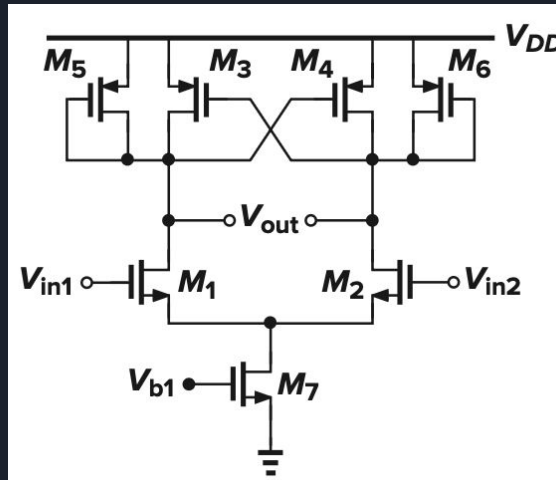
Problem 4.19

4.19. Consider the circuit shown in Fig. 4.46.

(a) Sketch V_{out} as V_{in1} and V_{in2} vary differentially from zero to V_{DD} .

(b) If $\lambda = 0$, obtain an expression for the voltage gain. What is the voltage gain if $W_{3,4} = 0.8W_{5,6}$?

Diagram -

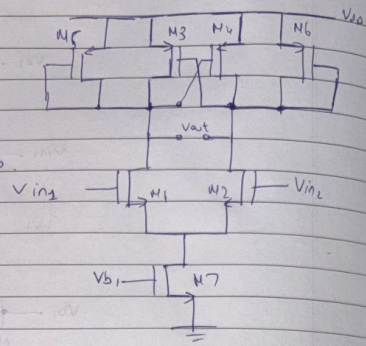


Handwritten Solution

4.19.7

$V_{in,CM} = V_{in1} = V_{in2}$ varies from 0 to V_{DD}

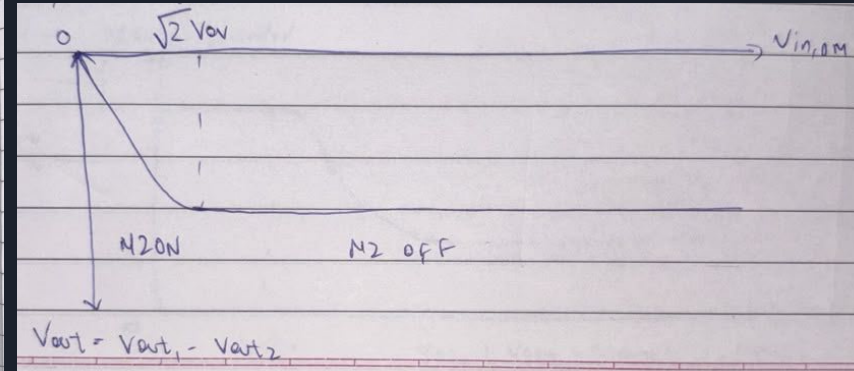
When $V_{in1} = V_{in2}$, i.e. $V_{in,CM} = 0$
 $V_{out1} = V_{out2}$, $\therefore V_{out} = 0$
 $V_{out} = V_{out1} - V_{out2}$



$\therefore$ As $V_{in,CM} \uparrow$, $I_{D1} \uparrow$, $I_{D2} \downarrow$
 $\therefore V_{out1} \downarrow$ and $V_{out2} \uparrow$, $\therefore V_{out} \downarrow$
 when $V_{out1} \downarrow$, it is the gate potential for M_4
 $V_{out1} = V_{G4}$, $\therefore M_4$ turns on stronger
 and pulls up V_{out2} , $\therefore V_{out} \uparrow$
 Similarly, $V_{out2} \uparrow$, $V_{G3} \uparrow$, $\therefore M_3$ strength is reduced which further reduces V_{out2}

$\therefore$ There is a positive feedback in the system

$\therefore$ As $V_{in,CM} \uparrow$, $V_{out} \downarrow$
 This is valid till M_2 is on, once M_2 turns off, V_{out} becomes constant



Handwritten Solution

b.4 $\therefore d=0$, Let V_{out1} increase by V_x , $\therefore V_{G4} \rightarrow V_{G4} + V_x$
 Let V_{out2} decrease by V_x , $\therefore V_{G3} \rightarrow V_{G3} - V_x$

$$I_x = I_{M3} = g_{m3} V_{G3} = g_{m3}(0 - V_{G3})$$

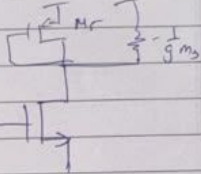
$$I_x = g_{m3} V_x$$

$\therefore$ looking into drain of $M3$

$$\therefore r \Rightarrow \frac{V_x}{-I_x} = - \frac{V_x}{g_{m3} V_x} = - \frac{1}{g_{m3}}$$

$\therefore$ cross coupled PMOS pair introduce -ve resistance

Gain



$$R_{out} = \frac{1}{g_{m1}} \parallel \frac{1}{g_{m3}}$$

$$A_v = -g_{m1} R_{out}$$

$$A_v = \frac{-g_{m1}}{g_{m1} - g_{m3}}$$

$\therefore$ If $g_{m1} - g_{m3}$ is very small gain can be increased

$$g_m = \mu_p C_{ox} \frac{W}{L} (V_{G4} - 1V_{thp1}) = \therefore 2 \mu_p C_{ox} \frac{W}{L} I_{D5}$$

$$\therefore I_{D3} + I_{D4} = I_{D5}, \quad I_{D3} = 0.8 I_{D5}$$

$$g_{m3} = \sqrt{2 \mu_p C_{ox} 0.8 \left(\frac{W}{L} \right) 0.8 I_{D5}}$$

$$g_{m3} = 0.8 g_{m1}$$

$$\therefore A_v = -5 \left(\frac{g_{m1}}{g_{m1} - g_{m3}} \right)$$

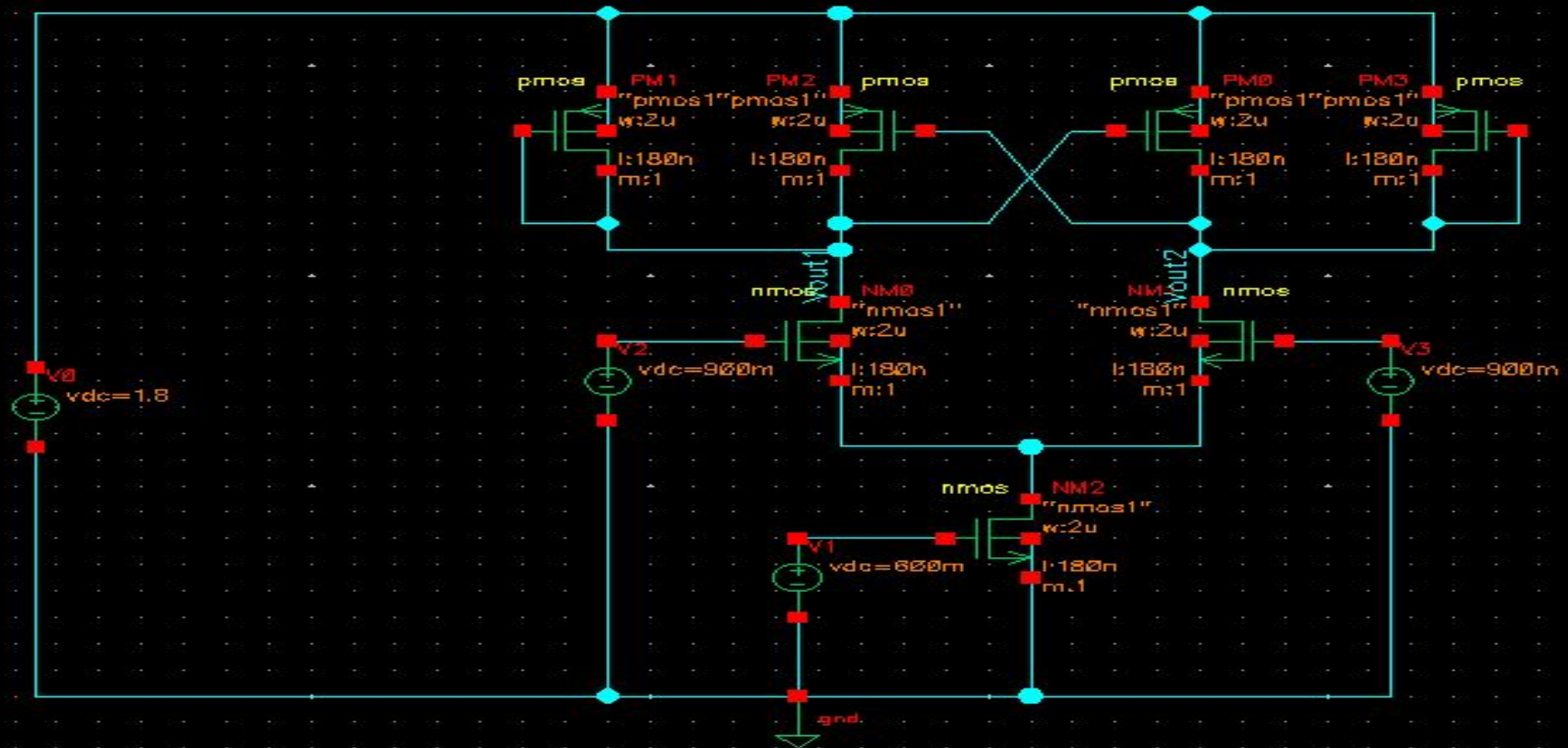


Cross-Coupled PMOS Pair: Gain Enhancement

- The cross-coupled PMOS pair (M3 and M4) introduces controlled positive feedback to increase the output resistance of the differential amplifier.
- The cross-coupling causes the PMOS pair to behave as an effective negative resistance: $R_{neg} = -1/gm_3$
- This negative resistance cancels the low resistance of the active load, giving: $R_{out} = 1/gm_5 - gm_3$
- When gm_3 is close to gm_5 , the output resistance increases significantly.
- Since: $A_v = -gm_1 R_{out}$, the differential voltage gain is greatly improved.
- The condition $gm_3 < gm_5$ is maintained to avoid excessive positive feedback and latch-up behavior.

Key idea: Cross-coupled PMOS loads achieve high gain by creating a high output impedance without requiring large resistors or extra voltage headroom.

Cadence Virtuoso Schematic



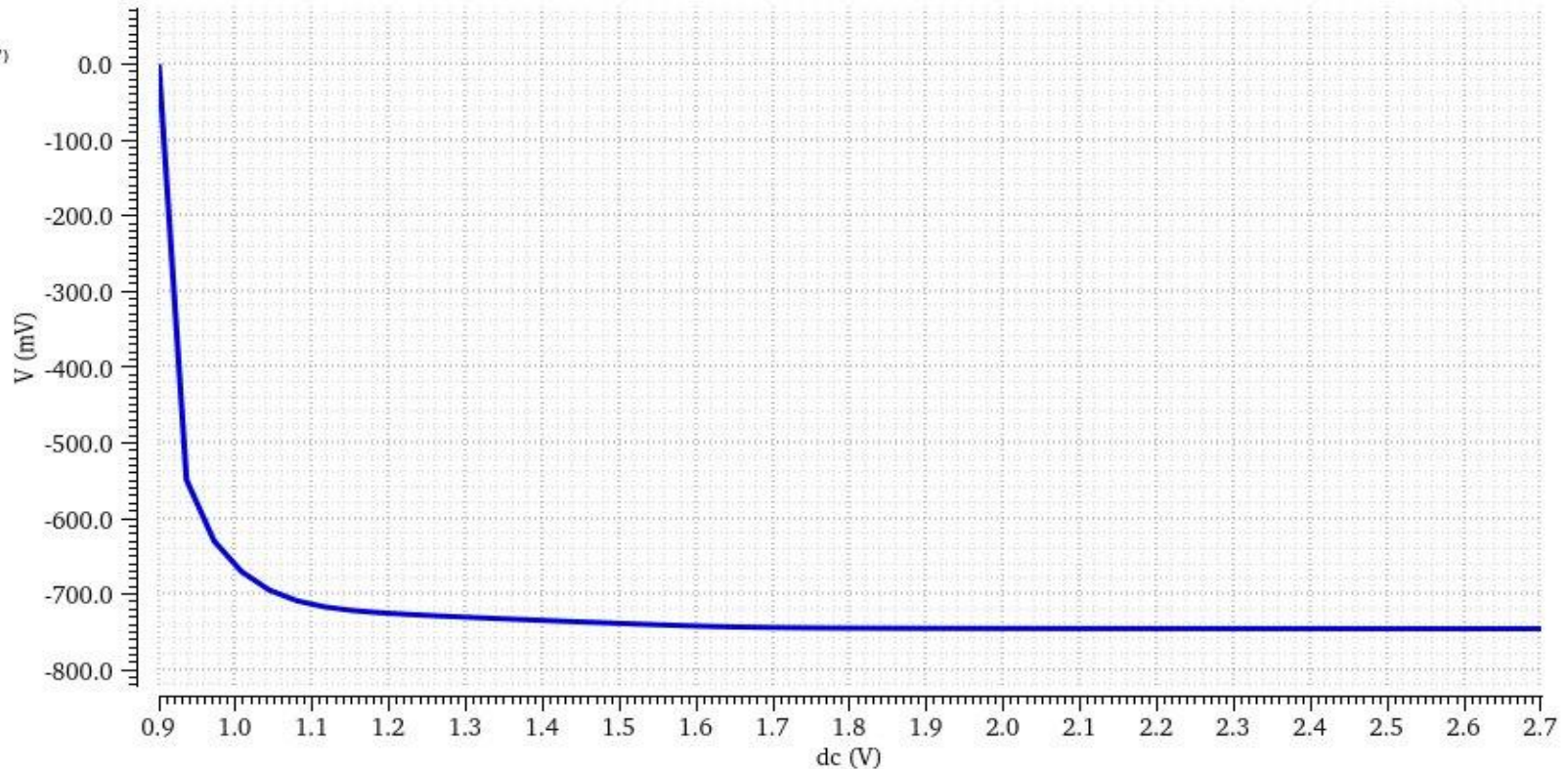
Cadence ADE-L DC Simulation for Vout

DC Response

Wed Jul 1 14:46:04 2026 1

Name

...-v("/Vout2" ?result "dc")

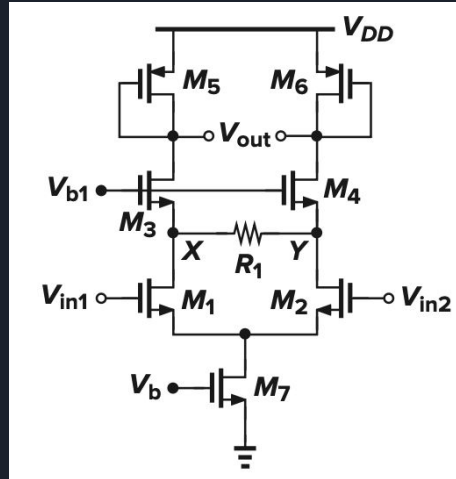


Problem 4.20

4.20. For the circuit shown in Fig. 4.47,

- (a) Sketch V_{out} , V_X , and V_Y as V_{in1} and V_{in2} vary differentially from zero to V_{DD} .
- (b) Calculate the small-signal differential voltage gain.

Diagram -

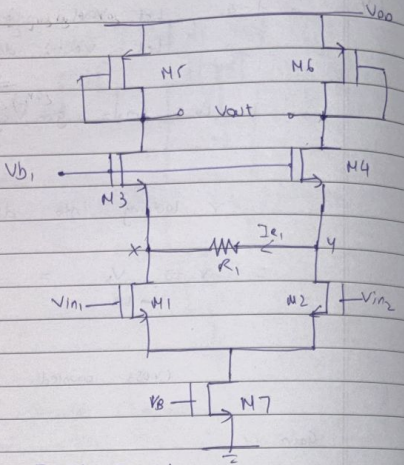


Handwritten Solution

4.20 y

a. y $V_{in,DM} = V_{in1} - V_{in2}$
 $V_{out} = V_{out1} - V_{out2}$
 $V_{in,DM}$ varies from 0 to V_{DD}

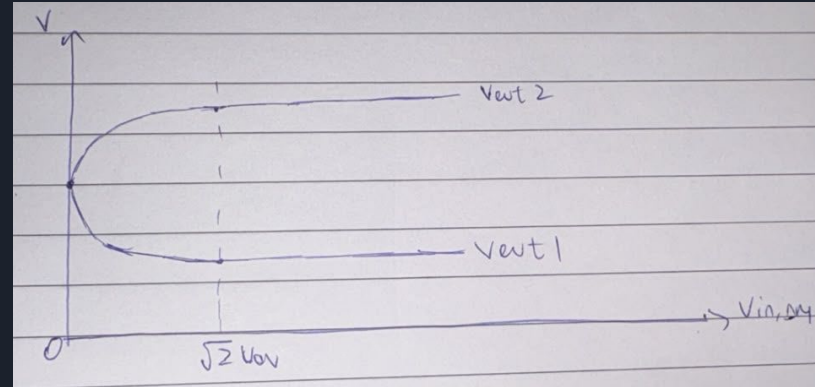
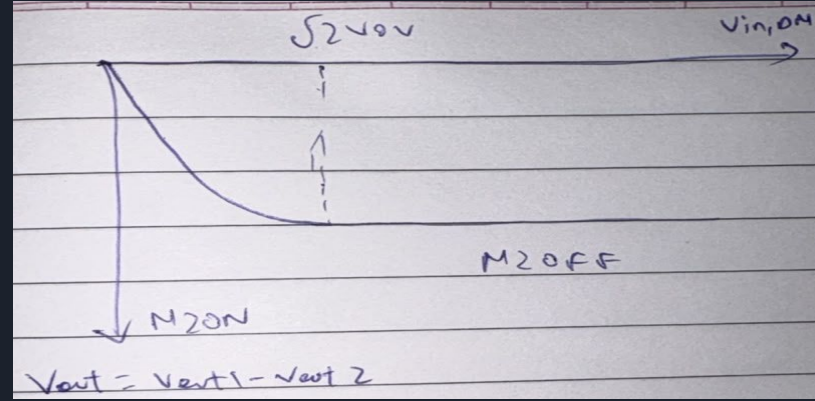
When $V_{in1} = V_{in2}$,
 $V_{out1} = V_{out2}$, $V_{out} = 0$
 $V_x = V_y$
 $I_{R1} = 0$



As $V_{in,DM}$ increases, I_{D1} , I_{D2} ↓↓
 V_{out1} ↓↓, V_{out2} ↑↑, $\therefore V_{out}$ ↓↓

Also, V_x ↓↓, V_y ↑↑, I_{R1} starts to flow from $y \rightarrow x$
 This is valid till $M2$ is ON

Once $M2$ turns off, entire current through $M6$ and $M4$ flows through $R1$ into $M1$, and
 $I_{M7} = I_{M1}$, $\therefore$ as $V_{in,DM}$ ↑↑, $V_y = \text{constant}$
 and $V_{out} = \text{constant}$
 $V_x = \text{constant}$
 $V_y > V_x$



Handwritten Solution

Points noted -

- V_X and V_Y vary linearly with $V_{in, dm}$ as the source nodes of the NMOS transistors see an effective resistance of approximately $1/g_m$.
- $G_M \neq g_{m1}$ because some current flows through $R_1/2$; the output current must be calculated using a current divider.

b.4 Gain

$$A_v = \frac{-g_m}{\frac{1}{g_{m5}} \parallel \left(\frac{v_{o1} R_1}{2} (1 + g_{m3} r_{o3}) + r_{o3} \right)}$$

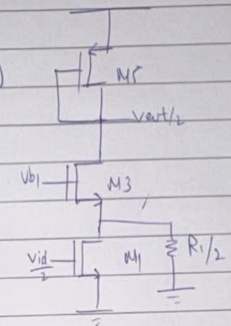
$$R_{out} = \frac{1}{g_{m5}} \parallel \left(\frac{R_1 g_{m3} r_{o3}}{2} \right)$$

$$R_{out} \approx \frac{1}{g_{m5}}$$

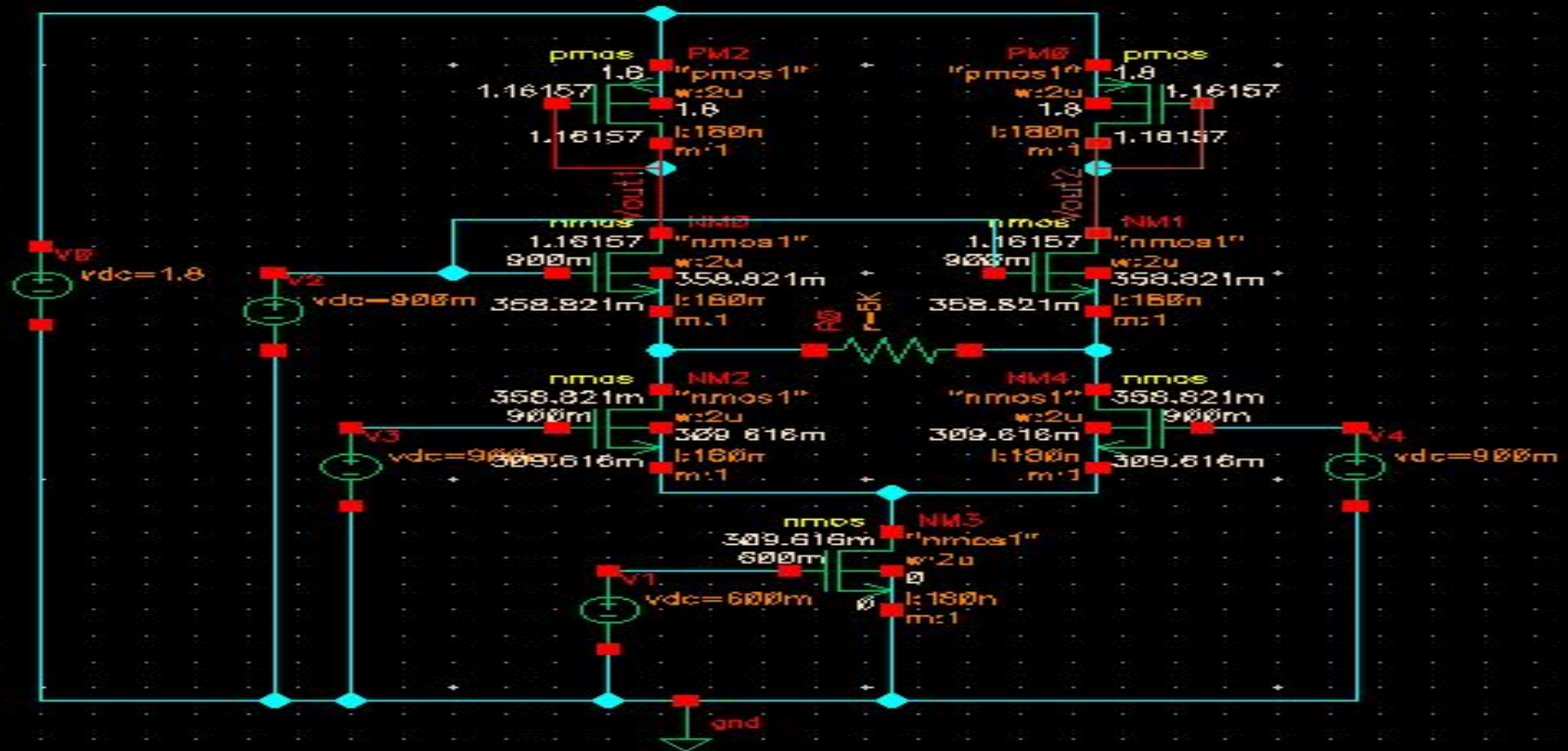
$$G_m = g_{m1} \left(\frac{\frac{R_1}{2}}{\frac{R_1}{2} + \frac{1}{g_{m3}}} \right)$$

Since current has 2 paths, we need the current going towards output

$$= g_{m1} \left(\frac{g_{m3} R_1}{2 + g_{m3} R_1} \right)$$

$$\therefore A_v = \frac{-g_{m1}}{g_{m5}} \left(\frac{g_{m3} R_1}{2 + g_{m3} R_1} \right)$$


Cadence Virtuoso Schematic



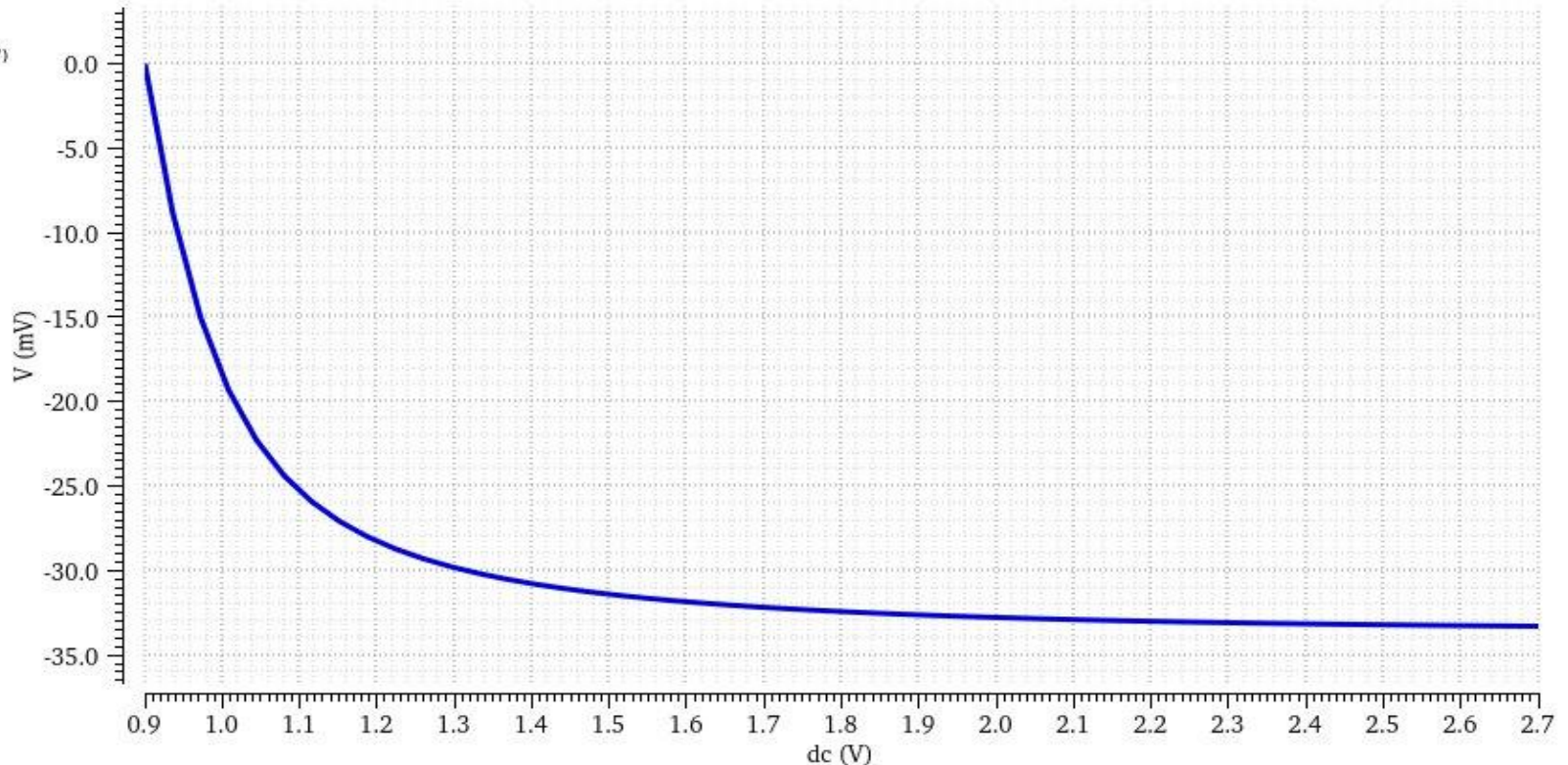
Cadence ADE-L DC Simulation for Vout1 - Vout2

DC Response

Wed Jul 1 14:26:30 2026 1

Name

....v("/Vout2" ?result "dc")



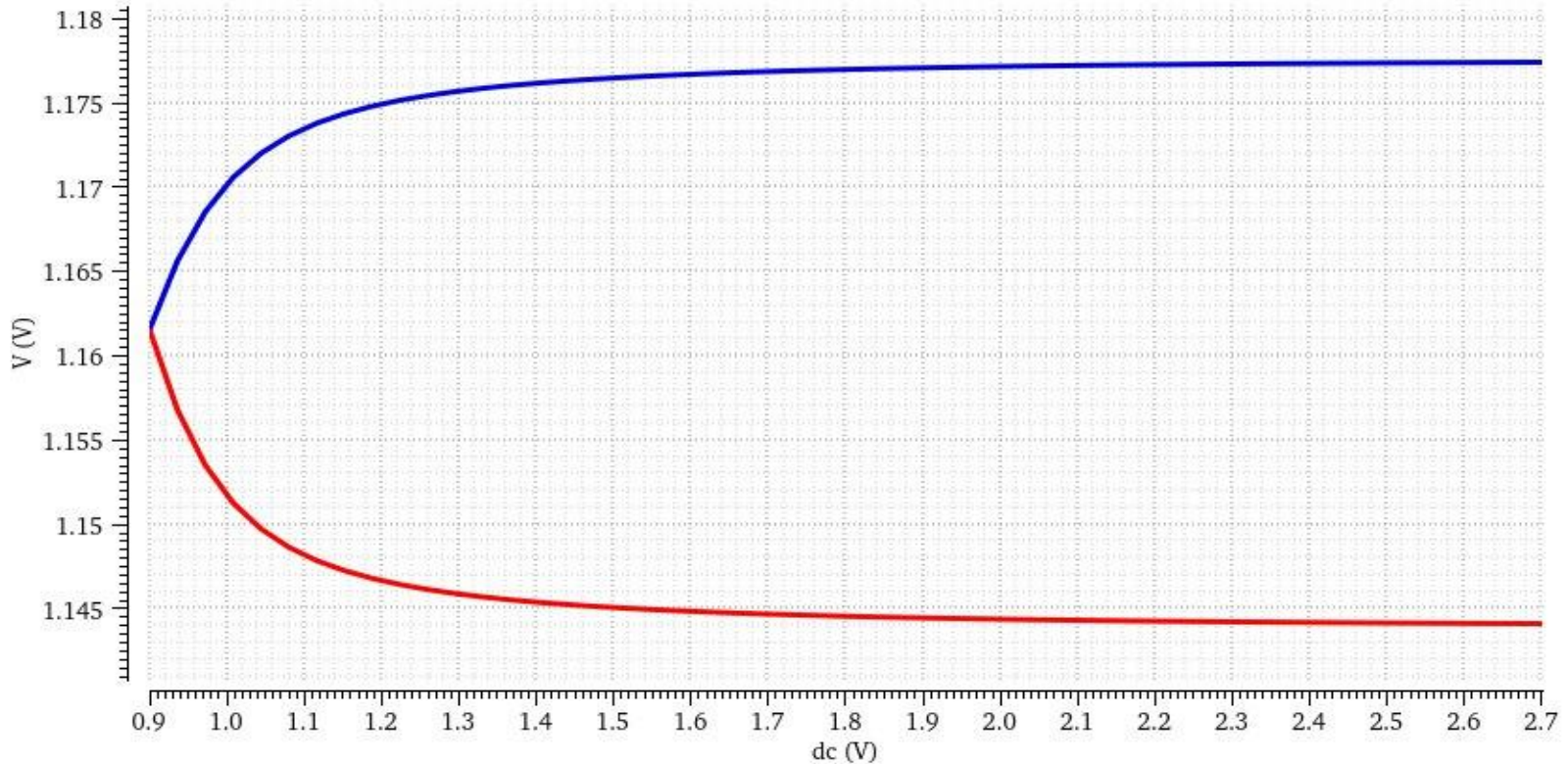
Cadence ADE-L DC Simulation for Vout

Wed Jul 1 14:26:30 2026 1

DC Response

Name	Vis
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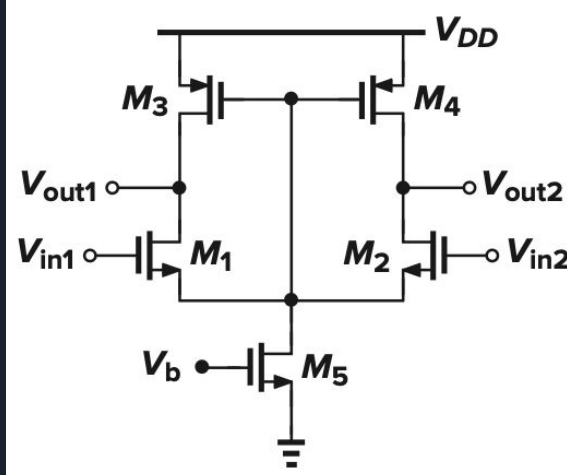
■ /Vout2	☒
■ /Vout1	☒



Problem 4.24

4.24. In the circuit of Fig. 4.51, all of the transistors have a W/L of 50/0.5, and M_3 and M_4 are to operate in the deep triode region with an on-resistance of 2 k Ω . Assuming that $I_{D5} = 20 \mu\text{A}$ and $\lambda = \gamma = 0$, calculate the input common-mode level that yields such resistance. Sketch V_{out1} and V_{out2} as V_{in1} and V_{in2} vary differentially from 0 to V_{DD} .

Diagram -



Handwritten Solution

4.25.4 $V_{SG3,4} = V_{DD} - V_x$

$R_{on} = \frac{1}{\mu_p C_{ox} \left(\frac{W}{L}\right)_3 (V_{ov})_3}$

$V_{ov3} = \frac{1}{\mu_p C_{ox} \frac{W}{L} R_{on}}$

$V_{ov3} = \frac{1}{60 \times 10^{-6} \times 100 \times 2 \times 10^3}$

$V_{ov3} = 0.0833 \text{ V}$

$V_{DD} - V_x - |V_{thp}| = 0.0833$

$1.8 - V_x - 0.5 = 0.0833$

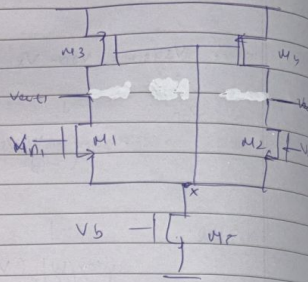
$V_x = 1.217 \text{ V}$

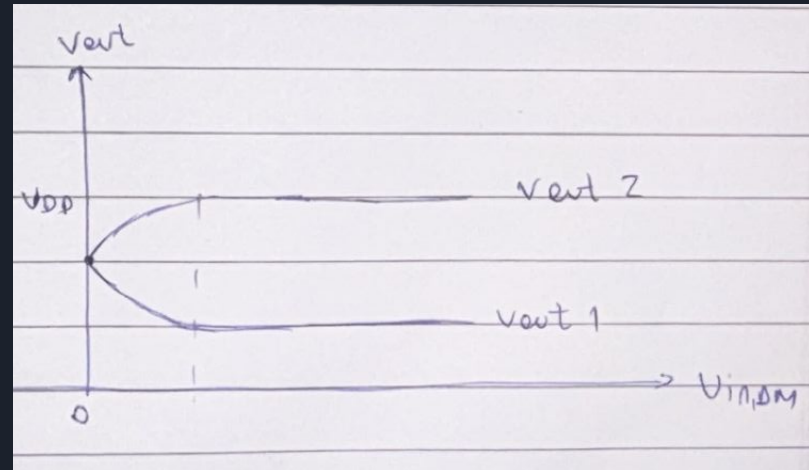
$I_{D1} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right)_1 (V_{in,cm} - V_x - V_{thn})^2 = 100 \mu\text{A}$

$10 \times 10^{-6} = \frac{1}{2} \times 300 \times 10^{-6} \times 100 \times (V_{in,cm} - 1.217 - 0.5)^2$

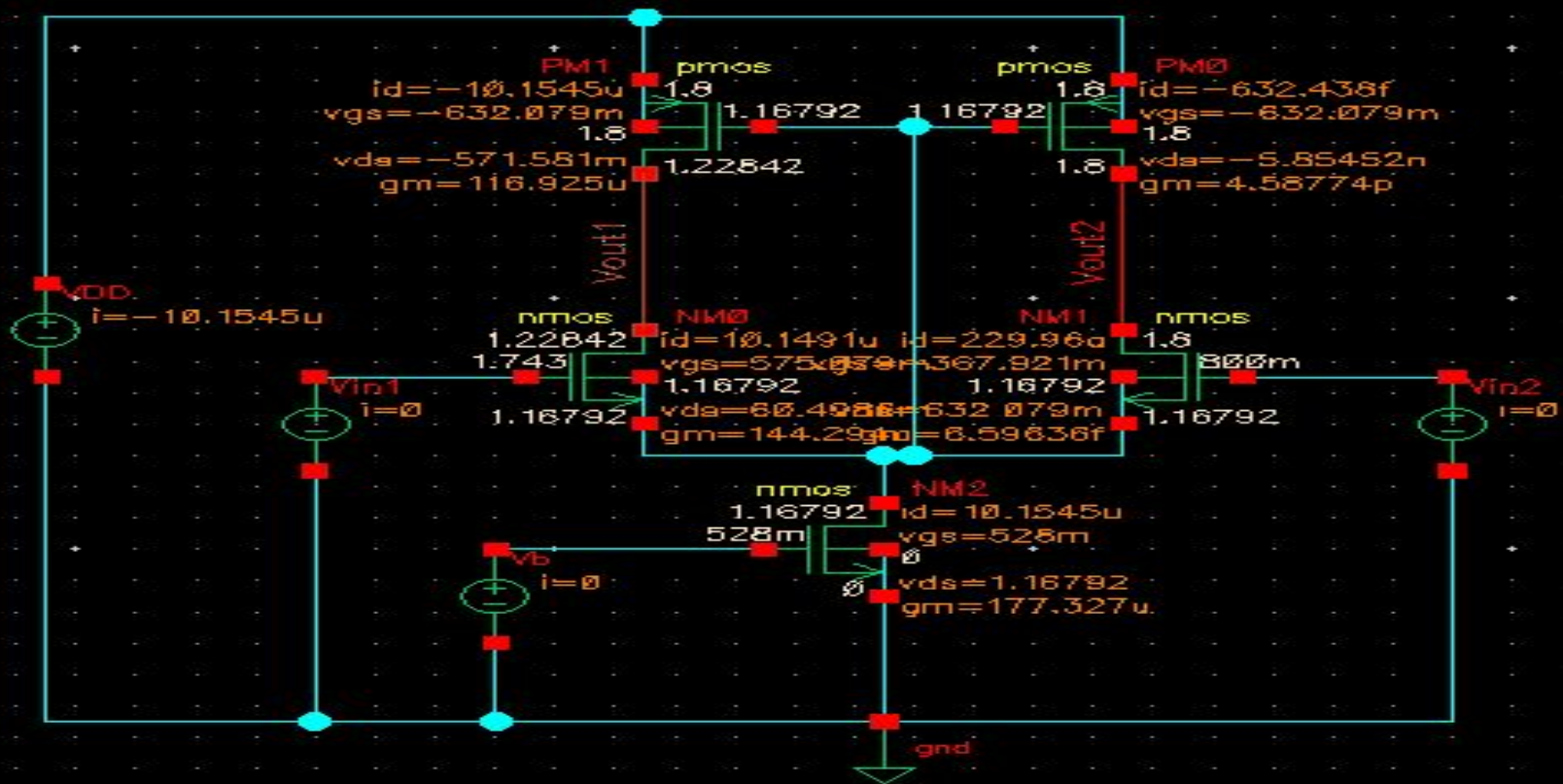
$6.667 \times 10^{-4} = (V_{in,cm} - 1.717)^2$

$V_{in,cm} = 1.743 \text{ V}$





Cadence Virtuoso Schematic



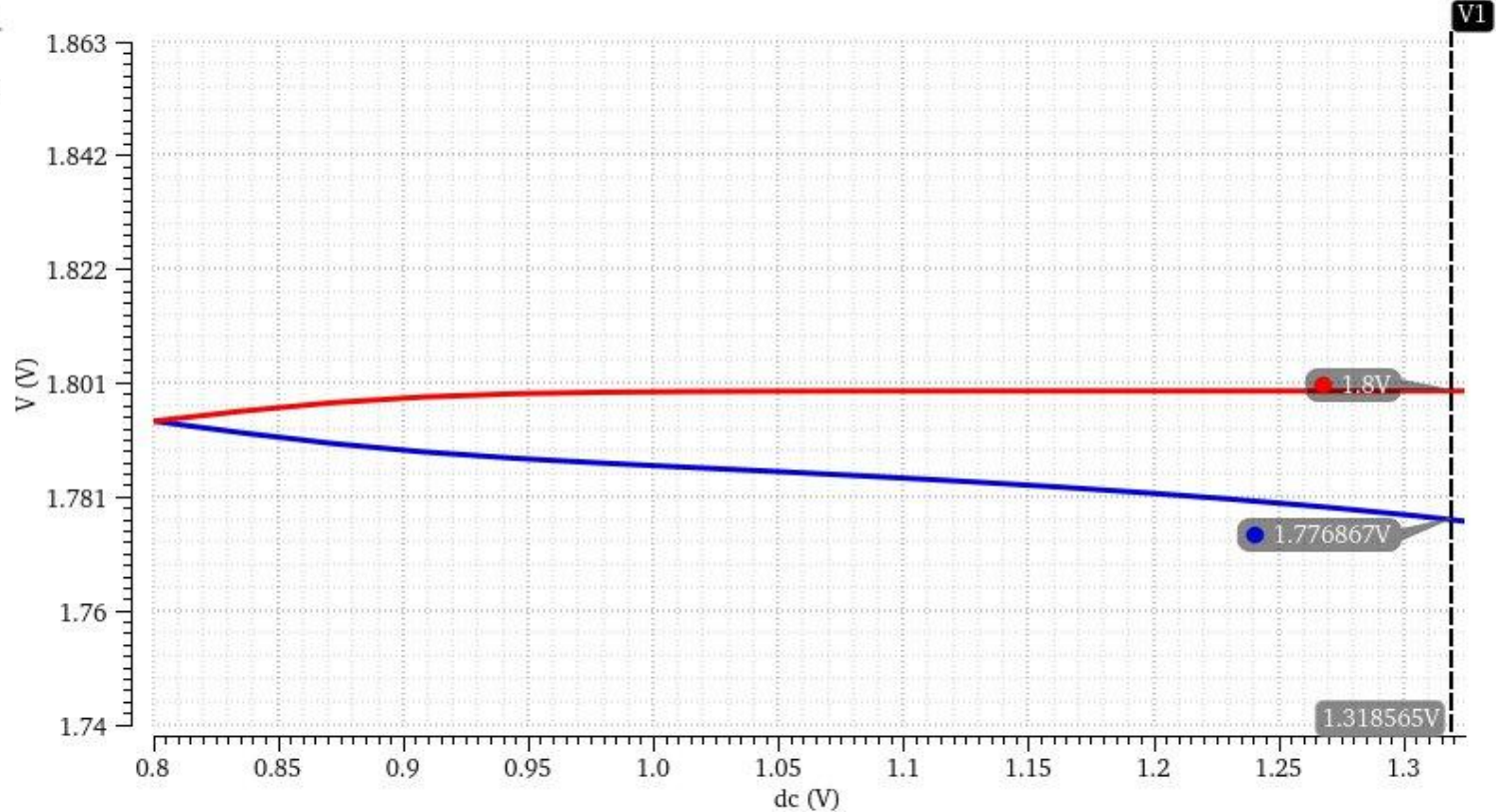
Cadence ADE-L DC Simulation for Vout

DC Response

Wed Jul 1 13:49:36 2026 1

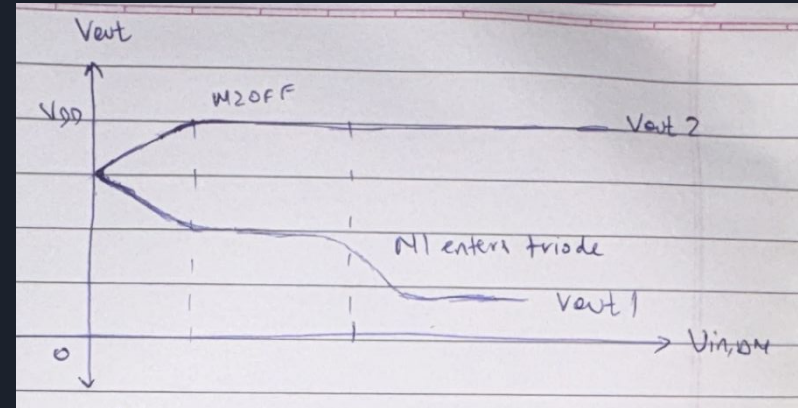
Name --- V1

■ /Vout1		1.776867V
■ /Vout2		1.8V



Points to be noted

- As V_{in1} increases, M1 draws more current, causing V_{out1} to decrease. The sharp drop occurs when M1 leaves saturation and enters triode, causing a rapid change in the output node voltage.
- V_{out2} remains close to V_{DD} because M2 is turned off, so the right branch carries negligible current.



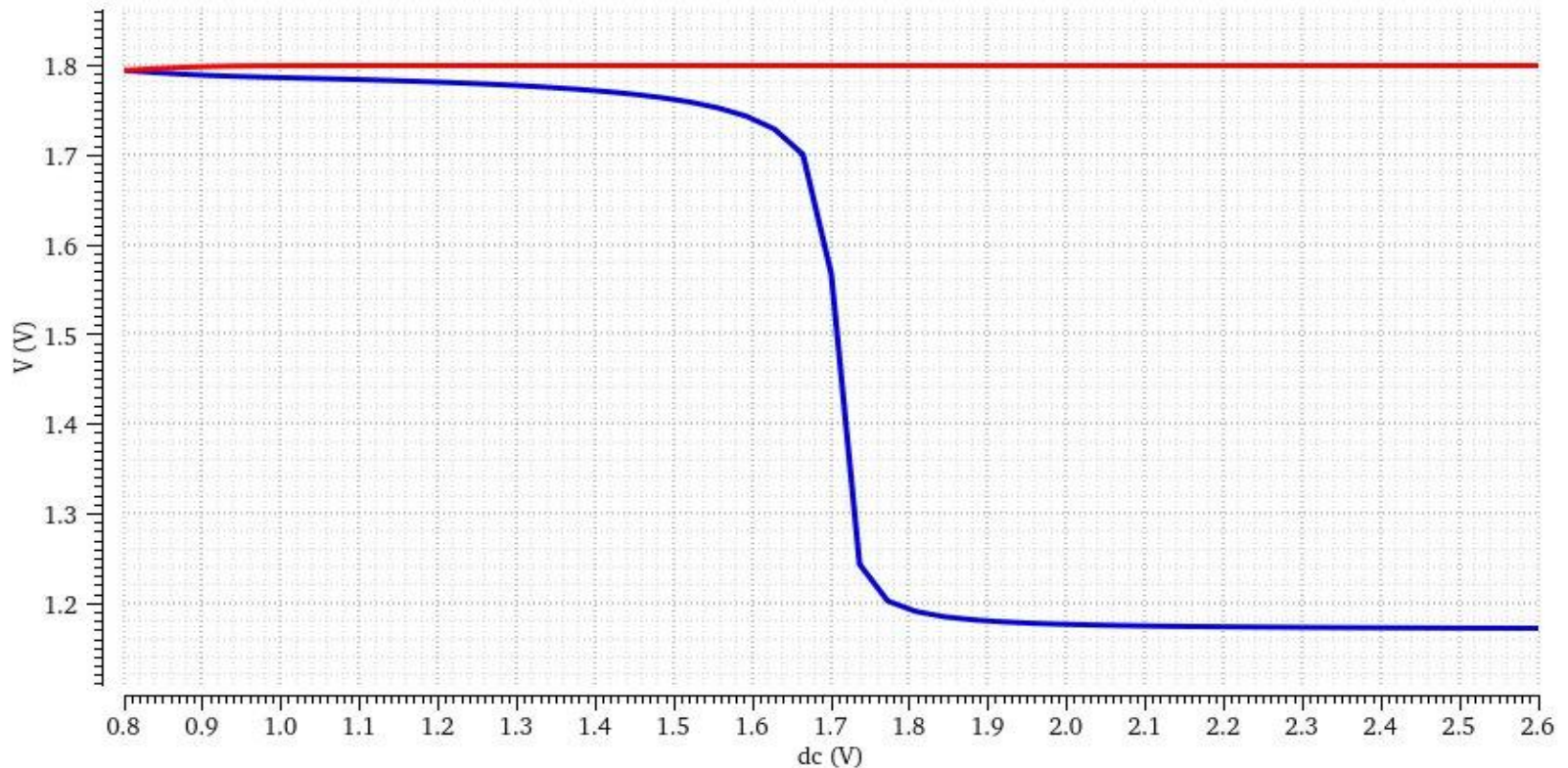
Cadence ADE-L DC Simulation for Vout

Wed Jul 1 13:49:36 2026 1

DC Response

Name	Vis
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■ /Vout1	
■ /Vout2	





THANK YOU